

Proposal of a viscoelastic-viscoplastic model for strain localization in geomaterials

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Plan

My plan is to study strain localization in a plasticity model using the tools of bifurcation theory in Banach spaces. Large strain models are too much difficult and until now I never found theorems ensuring sufficient smoothness in time to apply the tools of bifurcation theory. For that reason I looked for a strain localization phenomenon of mechanical interests also in a small deformation regime. In the field of metal plasticity the necking is a classical example of strain localization. Since Von Mises plasticity with hardening at small strain does not exhibit bifurcation (uniqueness is guaranteed) then to detect the necking instability it is necessary to consider the geometric nonlinearities making the problem very difficult. Since I am interested in geophysics I decided to study instead shear band formation in geomaterials or more in general with material with pressure dependent yield strength. I developed a new model by adapting the classical Drucker–Prager plasticity model to the framework of strain gradient plasticity of Gurtin and Anand based on the principle of virtual powers and on macroforce and microforce balances. Now I am carrying out the mathematical analysis of that model. Until now I have proved the local in time existence and uniqueness and the global in time existence of the discretized in time (this last existence result under the assumption of vanishing dilatancy). I plan to complete the proof of the global in time existence result and to carry a bifurcation analysis using the Lyapunov–Smith procedure and the Crandall–Rabinowitz theorem. Moreover I would like to confirm the mathematical analysis by a numerical study of the problem using the FEM. I would like to develop a method to detect bifurcation points during time evolution and to approximate the various branches of solutions that departs from them.

Chapter 1

Small deformation model

This chapter is devoted to the development of a mechanical model to describe the behavior of geomaterials in drained conditions. Even though the model is not derived from any particular micromechanical description of a geomaterial it is useful to keep in mind that the material is composed by interlocked grains that can rotate and deform.

1.1 Formulation

1.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^d$ be the reference configuration of the body. The macroscopic displacement is a map $u: \Omega \rightarrow \mathbb{R}^d$. We set $E = \text{sym grad } u$ and $M = \text{grad}^2 u$. We subdivide $\partial\Omega$ in two disjoint parts Γ_D and Γ_N . On Γ_D we impose the kinematical constrain $u = \bar{u}$. To impose this constrain we add the Lagrange multiplier $r: \Gamma_D \rightarrow \mathbb{R}^d$. The constrain can be written in weak form as

$$\int_{\Gamma_D} (u - \bar{u}) \cdot \tilde{r} da = 0 \quad \text{for all } \tilde{r}.$$

The motion of the grains of the geomaterial is described by the plastic distortion $H_p: \Omega \rightarrow \text{Lin}$. The skew symmetric part $W_p = \text{skew } H_p$ describes the rotation of the grains and for simplicity we assume that it vanish. The symmetric part is the plastic strain $E_p = \text{sym } H_p: \Omega \rightarrow \text{Sym}$ which describes the deformation of the grains. The volumetric deformation of the grains is described by $\lambda_p = \text{tr } E_p$ while the deviatoric deformation by $U_p = \text{dev } E_p$. We call $M_p = \text{grad } U_p$ and we set $\mathbb{U}_p = (U_p, l M_p) \in \text{DevPla}$ where $l > 0$ is an internal length scale reflecting the size of the grains. Here $\text{DevPla} = \text{Sym} \otimes (\text{Sym} \otimes \mathbb{R})$ is the vector space of the deviatoric plastic rates and their gradients. To describe the dilatant behavior of the geomaterial during plastic deformation we add the kinematical constrain

$$\dot{\lambda}_p = F(\lambda_p, \dot{U}_p)$$

where $F: \mathbb{R} \times \text{DevSym} \rightarrow \mathbb{R}$. To impose this constrain we add the Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$. The constrain can be written in weak form as

$$\int_{\Omega} (F(\lambda_p, \dot{U}_p) - \dot{\lambda}_p) \tilde{p} dx = 0 \quad \text{for all } \tilde{p}.$$

We can regard H_p as the elastically relaxed strain of the body so we define the elastic strain as $E_e = E - E_p$.

1.1.2 Statics

We choose an internal virtual power of the form

$$\begin{aligned}\tilde{W} &= \int_{\Omega} \left(\mathbb{T} : \tilde{\mathbb{E}}_e + \mathbb{K} : \tilde{\mathbb{M}} \right) dx + \int_{\Omega} \left(\mathbb{T}_p : \tilde{\mathbb{U}}_p + \mathbb{K}_p : \tilde{\mathbb{M}}_p \right) dx = \\ &= \int_{\Omega} \left(\mathbb{T} : \tilde{\mathbb{E}}_e + \mathbb{K} : \tilde{\mathbb{M}} \right) dx + \int_{\Omega} \mathbb{T}_p \circ \tilde{\mathbb{U}}_p dx.\end{aligned}$$

Here $\mathbb{T} \in \text{Sym}$ is the macrostress and $\mathbb{K} \in \text{Lin} \otimes \mathbb{R}$ is the macrohyperstress, $\mathbb{T}_p \in \text{DevSym}$ is the microforce and $\mathbb{K}_p \in \text{DevSym} \otimes \mathbb{R}$ is the microstress. We have used the notation $\mathbb{T}_p = (\mathbb{T}_p, l^{-1} \mathbb{K}_p)$. The symbol $\circ$ denotes the natural scalar product in DevPla . We assume an external virtual power of the form

$$\tilde{\mathcal{P}} = \int_{\Omega} \mathbf{f} \cdot \tilde{\mathbf{u}} dx + \int_{\Gamma_N} \mathbf{h} \cdot \tilde{\mathbf{u}} da.$$

and a reaction virtual power of the form

$$\tilde{\mathcal{R}} = \int_{\Gamma_D} \mathbf{r} \cdot \tilde{\mathbf{u}} da + \int_{\Omega} p \text{tr} \tilde{\mathbb{E}}_p dx.$$

Definition 1.1 (Principle of Virtual Powers). Equilibrium holds when $\tilde{W} = \tilde{\mathcal{P}} + \tilde{\mathcal{R}}$ for all virtual rates.

To get the local form of the equilibrium condition we apply the principle of virtual powers.

Remark. I still need to update this part since now I added the second gradient term.

Lemma 1.2. The local form of the equilibrium condition is

$$\begin{aligned}\text{div } \mathbb{T} + \mathbf{f} &= 0 \quad \text{in } \Omega \\ \mathbb{T} \mathbf{n} &= \mathbf{r} \quad \text{in } \Gamma_D \\ \mathbb{T} \mathbf{n} &= \mathbf{h} \quad \text{in } \Gamma_N \\ -\text{dev } \mathbb{T} - \text{div } \mathbb{K}_p + \mathbb{T}_p &= 0 \quad \text{in } \Omega \\ \frac{1}{d} \text{tr } \mathbb{T} + p &= 0 \quad \text{in } \Omega \\ \mathbb{K}_p \cdot \mathbf{n} &= 0 \quad \text{in } \partial\Omega.\end{aligned}$$

Proof. We first write the internal virtual power as

$$\begin{aligned}\tilde{W} &= - \int_{\Omega} \text{div } \mathbb{T} \cdot \tilde{\mathbf{u}} dx + \int_{\partial\Omega} \mathbb{T} \mathbf{n} \cdot \tilde{\mathbf{u}} da + \int_{\Omega} (-\text{dev } \mathbb{T} - \text{div } \mathbb{K}_p + \mathbb{T}_p) : \tilde{\mathbb{U}}_p dx - \\ &\quad - \int_{\Omega} \frac{1}{d} \text{tr } \mathbb{T} \text{tr} \tilde{\mathbb{E}}_p dx + \int_{\partial\Omega} (\mathbb{K}_p \cdot \mathbf{n}) : \tilde{\mathbb{U}}_p da.\end{aligned}$$

Then the results follows directly from the fundamental theorem of the calculus of variations. $\square$

1.1.3 Dissipation inequality

Let ψ be the free energy per unit volume in the reference configuration. Then the free energy contained in a portion Λ of Ω is

$$\Psi = \int_{\Lambda} \psi dx.$$

Let now $\mathcal{W}$ the internal power absorbed by the portion Λ . We define the dissipation power in Λ as

$$\mathcal{D} = \mathcal{W} - \dot{\Psi}.$$

The dissipation inequality impose that $\mathcal{D} \geq 0$ for all Λ .

1.1.4 Constitutive relations

We assume a constitutive relation for the free energy of the form

$$\psi = \frac{1}{2} \mathfrak{C} \mathbf{E}_e : \mathbf{E}_e + \frac{1}{2} \mathfrak{A} \mathbf{M} : \mathbf{M} + \frac{1}{2} \mathfrak{B} \mathbf{U}_p \circ \mathbf{U}_p.$$

with $\mathfrak{C}$, $\mathfrak{A}$ and $\mathfrak{B}$ a symmetric and positive definite linear operators on Sym , $\text{Lin} \otimes \mathbb{R}$ and DevPla respectively. To describe the viscoelastic response we use a constitutive relation of the form

$$\mathbf{T} = \mathfrak{C} \mathbf{E}_e + \mathfrak{D} \dot{\mathbf{E}}_e \quad \text{and} \quad \mathbf{K} = \mathfrak{A} \mathbf{M} + \mathfrak{E} \dot{\mathbf{M}}$$

with $\mathfrak{D}$ and $\mathfrak{E}$ symmetric and positive definite tensors on Sym and $\text{Lin} \otimes \mathbb{R}$ respectively. The viscoplastic response is obtained with a constitutive relation of the form

$$\mathbf{T}_p = \mathfrak{B} \mathbf{U}_p + \mathfrak{F} \dot{\mathbf{U}}_p + Y(p) \partial_{\dot{\mathbf{U}}_p} G(\dot{\mathbf{U}}_p)$$

where $\mathfrak{F}$ is a symmetric positive definite tensor on DevPla , $G: \text{DevSym} \rightarrow \mathbb{R}$ is the plastic dissipation potential satisfying $\partial_{\dot{\mathbf{E}}_p} G(\dot{\mathbf{E}}_p) : \dot{\mathbf{E}}_p \geq 0$ and $Y: \mathbb{R} \rightarrow (0, \infty)$ is the pressure dependent yield strength. We notice that using these constitutive relations the dissipation inequality is satisfied.

1.1.5 Refinements of the model

Until now the model is not able to describe compaction but only dilatancy so we could refine this aspect. We could add a field δ_p representing the volumetric deformation of the grains. Then the void level is represented by $\lambda_p - \delta_p$. In this case the constrain is $\dot{\lambda}_p - \dot{\delta}_p = F(\lambda_p - \delta_p, \dot{\mathbf{U}}_p)$. Now p would be power conjugate to $\tilde{\lambda}_p - \tilde{\delta}_p$. An additional microscopic force τ_p would appear power conjugate to $\tilde{\delta}_p$. We get the additional microforce balance

$$p + \tau_p = 0.$$

The constitutive relation could be $\tau_p = \partial \gamma_p(\tilde{\delta}_p)$ to describe a dissipative effect.

A further refinement could be the addition of plastic spin $\mathbf{W}_p \in \text{Skew}$. We notice that since the second gradient elasticity that we have introduced is independent of the plastic behavior¹ (this was done for mathematical reasons, since we needed compactness to apply Schauder theorem) then the rotational force would come only from the plastic behavior in the sense that: the macroscopic stress only exert a symmetric microforce that in principle only change $\mathbf{E}_p$ but the plastic response can cause also a nonsymmetric zero microforce so that $\mathbf{W}_p$ can became nonzero.

1.2 Mathematical analysis

In what follows we write simply $\mathbf{U}$ and λ in place of $\mathbf{U}_p$ and λ_p respectively.

¹Because $\mathbf{K}$ is conjugate to $\tilde{\mathbf{M}}$ and not to $\tilde{\mathbf{M}}_e$ with $\mathbf{M}_e = \text{grad } H_e$ and $H_e = H - H_p$)

1.2.1 Hypotheses

To get the global existence result we will assume the following hypotheses.

- (S) (a) $\Omega \subset \mathbb{R}^d$ with $d \in \{2, 3\}$ is open, bounded and Lipschitz
(b) Γ_D and Γ_N are relatively open disjoint subsets of $\partial\Omega$
- (F) (a) $F \in C^0(\mathbb{R} \times \text{DevSym}, [0, \infty))$
(b) there exists $\gamma_F \in C_c^{0,1}(\mathbb{R}, [0, \infty))$ such that $F(\lambda, \cdot)$ is Lipschitz in DevSym with Lipschitz constant $\gamma_F(\lambda)$
(c) $F(\cdot, 0) = 0$ and $F(\cdot, \dot{U})$ is nonincreasing for all $\dot{U} \in \text{DevSym}$
- (Y) (a) $Y \in C^{0,1}(\mathbb{R})$
(b) there exist $c_Y > 0$ such that $Y(p) \geq c_Y$ for all $p \in \mathbb{R}$
- (G) (a) $G \in C^{1,0}(\text{DevPla}, \mathbb{R})$ is convex
(b) $\partial G \in L^\infty(\text{DevPla}, \text{DevPla})$
(c) $\partial G(\dot{U}) : \dot{U} \geq 0$ for all $\dot{U} \in \text{DevPla}$
- (L) (a) $f \in C^1([0, T], L^2(\Omega)^d)$
(b) $h \in C^1([0, T], L^2(\Gamma_N)^d)$
- (E) (a) $\mathfrak{C}$ and $\mathfrak{D}$ are in $\text{Sym} \otimes \text{Sym}$, $\mathfrak{B}$ and $\mathfrak{F}$ are in $\text{DevPla} \otimes \text{DevPla}$, $\mathfrak{A}$ and $\mathfrak{E}$ are in $(\text{Lin} \otimes \mathbb{R}) \otimes (\text{Lin} \otimes \mathbb{R})$ and $\mathfrak{F}$ is in $\text{DevPla} \otimes \text{DevPla}$
(b) there exist $\gamma > 0$ and $\delta > 0$ such that $\mathfrak{C}E : E \geq \gamma|E|^2$ and $\mathfrak{D}E : E \geq \delta|E|^2$ for all $E \in \text{Sym}$, there exist $\beta > 0$ and ϕ such that $\mathfrak{B}U \circ U \geq \beta|U|^2$ and $\mathfrak{F}U \circ U \geq \phi|U|^2$, there exists $\alpha > 0$ and $\epsilon > 0$ such that $\mathfrak{A}M : M \geq \alpha|\text{sym } M|^2$ and $\mathfrak{E}M : M \geq \epsilon|\text{sym } M|^2$ for all $M \in \text{Lin} \otimes \mathbb{R}$ where sym denote the symmetrization of the first two indices

The local existence and uniqueness theorem and the bifurcation analysis is made under the following additional hypotheses.

- (F) (d) $F \in C^{1,1}(\mathbb{R} \times \text{DevSym}, [0, \infty))$
(e) $\partial_{\dot{U}} F(\lambda, 0) = 0$ for all $\lambda \in \mathbb{R}$
- (Y) (a) $Y \in C^1(\mathbb{R})$
- (G) (d) $G \in C^2(\text{DevPla})$ and $\partial^2 G$ is bounded
(e) there exists $\gamma_G : \text{DevPla} \rightarrow (0, \infty)$ such that $\partial^2 G(\dot{U})\mathbb{V} \circ \mathbb{V} \geq \gamma_G(\dot{U})|\mathbb{V}|^2$ for all $\dot{U}$ and $\mathbb{V}$ in DevPla
(f) $\partial G(0) = 0$

1.2.2 Function spaces

We now introduce the function spaces that we use to formulate the problem. Let

$$\mathcal{Z} = \left\{ y = (u, U, \lambda) \left| \begin{array}{l} u \in H^2(\Omega)^d \text{ and } u = 0 \text{ on } \Gamma_D \\ U \in H^1(\Omega, \text{DevSym}) \\ \lambda \in L^2(\Omega) \end{array} \right. \right\} \quad \text{and} \quad \mathcal{Q} = \{ p \in L^2(\Omega) \}.$$

The rates of $y = (u, U, \lambda) \in \mathcal{Z}$ will be denoted by $\dot{y} = (\dot{u}, \dot{U}, \dot{\lambda}) \in \mathcal{Z}$. The variations of $\dot{y} = (\dot{u}, \dot{\xi}, \dot{\lambda}) \in \mathcal{Z}$ and $p \in \mathcal{Q}$ will be denoted as $z = (v, V, \mu) \in \mathcal{Z}$ and $q \in \mathcal{Q}$. We define also the spaces of test functions as

$$\tilde{\mathcal{Z}} = \left\{ \tilde{z} = (\tilde{v}, \tilde{V}, \tilde{\mu}) \left| \begin{array}{l} \tilde{v} \in H^2(\Omega)^d \text{ and } \tilde{v} = 0 \text{ on } \Gamma_D \\ \tilde{V} \in H^1(\Omega, \text{DevSym}) \\ \tilde{\mu} \in L^2(\Omega) \end{array} \right. \right\} \quad \text{and} \quad \tilde{\mathcal{Q}} = \{ \tilde{q} \in L^2(\Omega) \}.$$

1.2.3 Formulation of the problem

We define $\mathcal{F}: \mathcal{Z} \times \mathcal{Q} \times \mathcal{Z} \times [0, T] \rightarrow \tilde{\mathcal{Z}}' \times \tilde{\mathcal{Q}}'$ by

$$\begin{aligned} \langle \mathcal{F}(\dot{y}, p, y, t), (\tilde{z}, \tilde{q}) \rangle &= \int_{\Omega} \left[\mathfrak{E} \left(\mathbb{E}u - u - \frac{\lambda}{d} I \right) + \mathfrak{D} \left(\mathbb{E}\dot{u} - \dot{u} - \frac{\dot{\lambda}}{d} I \right) \right] : \left(\mathbb{E}\tilde{v} - \tilde{v} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &+ \int_{\Omega} (\mathfrak{A} \operatorname{grad}^2 u + \mathfrak{E} \operatorname{grad}^2 \dot{u}) : \operatorname{grad}^2 \tilde{v} dx + \int_{\Omega} [(\mathfrak{B}U + \mathfrak{F}\dot{U} \circ \tilde{V} + Y(p)\partial G(\dot{U}) \circ \tilde{V})] dx - \\ &- \int_{\Omega} p \tilde{\mu} dx + \int_{\Omega} (F(\lambda, \dot{u}) - \dot{\lambda}) \tilde{q} dx - \\ &- \int_{\Omega} f(t) \cdot \tilde{v} dx - \int_{\Gamma_N} h(t) \cdot \tilde{v} dx. \end{aligned}$$

Lemma 1.3. The map $\mathcal{F}$ is well defined.

Proof. The linear terms are trivially well defined so we need to check only

$$\int_{\Omega} Y(p)\partial G(\dot{U}) \circ \tilde{V} dx \quad \text{and} \quad \int_{\Omega} F(\lambda, \dot{u}) \tilde{q} dx. \quad (1.2.1)$$

But

$$\int_{\Omega} |Y(p)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \|\partial G\|_{\infty} \int_{\Omega} (|Y(p)| + \operatorname{Lip}(Y)|p|) |\tilde{V}| dx \lesssim (1 + \|p\|_2) \|\tilde{V}\|_{1,2}$$

and

$$\int_{\Omega} |F(\lambda, \dot{u}) - \dot{\lambda}| |\tilde{q}| dx \leq \int_{\Omega} [\|\gamma_F\|_{\infty} |\dot{u}| + |\dot{\lambda}|] |\tilde{q}| dx \lesssim (\|\dot{u}\|_2 + \|\dot{\lambda}\|_2) \|\tilde{q}\|_2.$$

This concludes the proof. $\square$

Now we can formulate the evolution problem as follows.

Problem 1.4. Find $(y, p) \in AC([0, T], \mathcal{Z} \times \mathcal{Q})$ (maybe we will need to change this space, for example maybe the time regularity of p can be worser) such that $y(0) = 0$ and

$$\mathcal{F}(\dot{y}(t), p(t), y(t), t) = 0$$

for almost every $t \in (0, T)$.

1.2.4 Global existence theorem

Remark. We plan to use the Rothe method. Until now I have proved the existence of the time discrete solution. Now I am trying to pass to the time continuous limit in the case of no dilatancy ($F = 0$).

Let $N \in \mathbb{N}$. We define $\tau = N^{-1}T$ and $t_n = n\tau$ for $n \in \{0, \dots, N\}$. We consider the discrete in time version of Problem 1.4.

Problem 1.5 (Discrete problem). Let $y_0 = 0 \in \mathcal{Z}$. Find $y_n \in \mathcal{Z}$ and $p_n \in \mathcal{Q}$ for $n \in \{1, \dots, N\}$ such that

$$\mathcal{F}\left(\frac{y_n - y_{n-1}}{\tau}, p_n, y_n, t_n\right) = 0 \quad (1.2.2)$$

for all $n \in \{1, \dots, N\}$.

We first prove the existence of a solution to Problem 1.5. Reasoning by induction we reduce to find $y_n \in \mathcal{Z}$ and $p_n \in \mathcal{Q}$ that satisfy (1.2.2) given $y_{n-1} \in \mathcal{Z}$. We fix any $t \in [0, T - \tau]$ and $y \in \mathcal{Z}$. It is enough to prove that there exists $\dot{y} \in \mathcal{Z}$ and $p \in \mathcal{Q}$ such that

$$\mathcal{F}(\dot{y}, p, y + h\dot{y}, t) = 0. \quad (1.2.3)$$

We define the operator $\mathcal{A}: \mathcal{Q} \times \mathcal{Z} \rightarrow \tilde{\mathcal{Z}}'$ by

$$\begin{aligned} \langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle &= \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{E}) \left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \operatorname{grad}^2 \dot{u} : \operatorname{grad}^2 \tilde{v} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \tilde{V} + Y(p) \partial G(\dot{U}) \circ \tilde{V}] dx \end{aligned}$$

the linear operator $\mathcal{B}: \mathcal{Q} \rightarrow \tilde{\mathcal{Z}}'$ by

$$\langle \mathcal{B}p, \tilde{z} \rangle = - \int_{\Omega} p \tilde{\mu} dx$$

and the operator $\mathcal{C}: \mathcal{Z} \rightarrow \tilde{\mathcal{Q}}'$ by

$$\langle \mathcal{C}(\dot{y}), \tilde{q} \rangle = \int_{\Omega} (F(\lambda + \tau \dot{\lambda}, \dot{U}) - \dot{\lambda}) \tilde{q} dx = 0.$$

Moreover we introduce the linear form $l \in \tilde{\mathcal{Z}}'$

$$\begin{aligned} \langle l, \tilde{z} \rangle &= \int_{\Omega} f(t) \cdot \tilde{v} dx + \int_{\Gamma_N} h(t) \cdot \tilde{v} dx - \\ &- \int_{\Omega} \mathfrak{E} \left(E\dot{u} - \dot{U} - \frac{\dot{\lambda}}{d} I \right) : \left(E\tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx - \\ &- \int_{\Omega} \mathfrak{A} \operatorname{grad}^2 \dot{u} : \operatorname{grad}^2 \tilde{v} dx - \int_{\Omega} \mathfrak{B} \dot{U} \circ \tilde{V} dx. \end{aligned}$$

Now equation (1.2.3) can be written as

$$\begin{cases} \mathcal{A}(p, \dot{y}) + \mathcal{B}p = l(t) \\ \mathcal{C}(\dot{y}) = 0. \end{cases}$$

We first solve the second equation.

Lemma 1.6. There exists $\Phi \in C^{0,1}(\operatorname{DevSym}) \cap L^{\infty}(\operatorname{DevSym})$ such that $\dot{y} \in \mathcal{Z}$ solves $\mathcal{C}(\dot{y}) = 0$ if and only if $\dot{\lambda} = \Phi(\dot{U})$. Moreover $\operatorname{Lip}(\Phi) \leq \|\gamma_F\|_{\infty}$.

Proof. By hypotheses (F, a) for each $\dot{U} \in \operatorname{DevSym}$ the function $\dot{\lambda} \mapsto F(\lambda + \tau \dot{\lambda}, \dot{U})$ is continuous, strictly increasing and tends to $\pm\infty$ when $\dot{\lambda} \rightarrow \pm\infty$. Then there exists a unique $\Phi(\dot{U}) \in [0, \infty)$ such that $\Phi(\dot{U}) = F(\lambda + \tau \Phi(\dot{U}), \dot{U})$. Moreover

$$\Phi(\dot{U}) \leq F(\Phi(\dot{U}), 0) + \gamma_F(\lambda + \tau \Phi(\dot{U})) |\dot{U}| \leq \gamma_F(\lambda + \tau \Phi(\dot{U})) |\dot{U}|.$$

Since γ_F has compact support then $\Phi \in L^{\infty}$. This implies that given $\dot{U} \in H^1$ we have $\Phi(\dot{U}) \in L^{\infty} \subset L^2$. Then $\mathcal{C}(\dot{y}) = 0$ if and only if $\dot{\lambda} = \Phi(\dot{U})$. We now prove that Φ is Lipschitz. Let $\dot{\lambda}_2 = \Phi(\dot{U}_2)$ and $\dot{\lambda}_1 = \Phi(\dot{U}_1)$. Without loss of generality assume that $\dot{\lambda}_2 > \dot{\lambda}_1$. Then we have

$$\begin{aligned} \dot{\lambda}_2 - \dot{\lambda}_1 &= F(\lambda + \tau \dot{\lambda}_2, \dot{U}_2) - F(\lambda + \tau \dot{\lambda}_1, \dot{U}_1) = \\ &= F(\lambda + \tau \dot{\lambda}_2, \dot{U}_2) - F(\lambda + \tau \dot{\lambda}_1, \dot{U}_2) + F(\lambda + \tau \dot{\lambda}_1, \dot{U}_2) - F(\lambda + \tau \dot{\lambda}_1, \dot{U}_1) \leq \\ &\leq F(\lambda + \tau \dot{\lambda}_1, \dot{U}_2) - F(\lambda + \tau \dot{\lambda}_1, \dot{U}_1) \leq \|\gamma_F\|_{\infty} |\dot{U}_2 - \dot{U}_1| \end{aligned}$$

and

$$\begin{aligned}\dot{\lambda}_1 - \dot{\lambda}_2 &= F(\lambda + \tau\dot{\lambda}_2, \dot{U}_2) - F(\lambda + \tau\dot{\lambda}_1, \dot{U}_1) = \\ &= F(\lambda + \tau\dot{\lambda}_2, \dot{U}_2) - F(\lambda + \tau\dot{\lambda}_2, \dot{U}_1) + F(\lambda + \tau\dot{\lambda}_2, \dot{U}_1) - F(\lambda + \tau\dot{\lambda}_1, \dot{U}_1) \leq \\ &\leq F(\lambda + \tau\dot{\lambda}_2, \dot{U}_2) - F(\lambda + \tau\dot{\lambda}_2, \dot{U}_1) \leq \|\gamma_F\|_\infty |\dot{U}_2 - \dot{U}_1|.\end{aligned}$$

This concludes the proof. $\square$

We call

$$\mathcal{N} = \left\{ \dot{y} = (\dot{u}, \dot{U}, \Phi(\dot{U})) \mid \begin{array}{l} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}$$

and

$$\tilde{\mathcal{N}} = \{ \tilde{z} \in \tilde{\mathcal{Z}} \mid \langle \mathcal{B}q, \tilde{z} \rangle = 0 \text{ for all } q \in Q \} = \{ \tilde{z} \in \tilde{\mathcal{Z}} \mid \tilde{\mu} = 0 \}.$$

Now we study the first equation in (1.2.3). We first study the solvability of that equation in $\dot{y} \in \mathcal{N}$ given $p \in \mathcal{Q}$. If we test the first equation with $\tilde{z} \in \tilde{\mathcal{N}}$ we get

$$\langle \mathcal{A}(p, \dot{y}), \tilde{z} \rangle = \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{\mathcal{N}}.$$

Now we define the space

$$\mathcal{X} = \left\{ \xi = (\dot{u}, \dot{U}) \mid \begin{array}{l} \dot{u} \in H^2(\Omega)^d \text{ and } \dot{u} = 0 \text{ on } \Gamma_D \\ \dot{U} \in H^1(\Omega, \text{DevSym}) \end{array} \right\}$$

and we can consider the natural bijections $\mathcal{J}$ and $\tilde{\mathcal{J}}$ from $\mathcal{X}$ to $\mathcal{N}$ and $\tilde{\mathcal{N}}$ respectively. In particular if $\xi = (\dot{u}, \dot{U})$ and $\tilde{\xi} = (\tilde{v}, \tilde{V})$ are in $\mathcal{X}$ we define

$$\mathcal{J}(\xi) = (\dot{u}, \dot{U}, \Phi(\dot{U})) \quad \text{and} \quad \tilde{\mathcal{J}}\tilde{\xi} = (\tilde{v}, \tilde{V}, 0).$$

Now we introduce the operator $\mathcal{T}_p$ from $\mathcal{X}$ to $\mathcal{X}'$ as follows. Given $\xi = (\dot{u}, \dot{U})$ and $\tilde{\xi} = (\tilde{v}, \tilde{V})$ in $\mathcal{X}$ we set

$$\begin{aligned}\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle &= \langle \mathcal{A}(p, \mathcal{J}(\xi)), \tilde{\mathcal{J}}(\tilde{\xi}) \rangle = \int_{\Omega} (\mathfrak{D} + \tau\mathfrak{E}) \left(E\dot{u} - \dot{U} - \frac{\Phi(\dot{U})}{d} I \right) : (E\tilde{v} - \tilde{V}) \, dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau\mathfrak{A}) \text{grad}^2 \dot{u} : \text{grad}^2 \tilde{v} \, dx + \int_{\Omega} [(\tilde{\mathfrak{F}} + \tau\mathfrak{B})\dot{U} \circ \tilde{V} + Y(p)\partial G(\dot{U}) \circ \tilde{V}] \, dx.\end{aligned}$$

Now for each $p \in \mathcal{Q}$ we are reduced to find $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle = \langle l, \tilde{\mathcal{J}}\tilde{\xi} \rangle \quad \text{for all } \tilde{\xi} \in \tilde{\mathcal{X}}. \quad (1.2.4)$$

We want to apply the Browder–Minty theorem (Theorem B.15). We need to establish the boundedness, continuity, coercivity and monotonicity of $\mathcal{T}_p$. This done in the following lemmas.

Lemma 1.7. The operator $\mathcal{T}_p : \mathcal{X} \rightarrow \mathcal{X}'$ is bounded and continuous.

Proof. By Lemma 1.6 we know that if $\dot{U}_n \rightarrow \dot{U}$ in H^1 then

$$\|\Phi(\dot{U}_n) - \Phi(\dot{U})\|_2 \leq \text{Lip}(\Phi) \|\dot{U}_n - \dot{U}\|_2 \rightarrow 0.$$

Moreover since $\Phi \in L^\infty$ we have that $\mathcal{T}_p$ is bounded. Now

$$\begin{aligned}|\langle \mathcal{T}_p(\xi), \tilde{\xi} \rangle| &\leq (\|E\dot{u}\|_2 + \|\dot{U}\|_2 + \|\Phi\|_\infty) (\|E\tilde{v}\|_2 + \|\tilde{V}\|_2) + \\ &+ \|\text{grad}^2 \dot{u}\|_2 \|\text{grad}^2 \tilde{v}\|_2 + \|\dot{U}\|_2 \|\tilde{V}\|_2 + \|p\|_2 \|\tilde{V}\|_2.\end{aligned}$$

and this proves that $\mathcal{T}_p$ is bounded. Now it is enough to prove the continuity of the nonlinear term only. Let $\xi_n \rightarrow \xi$ in $\mathcal{X}$. Then $\dot{\mathbb{U}}_n \rightarrow \dot{\mathbb{U}}$ in L^2 . We have

$$\left| \int_{\Omega} Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}})) \circ \tilde{\mathbb{V}} dx \right| \leq \|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2 \|\tilde{\mathbb{V}}\|_2$$

and

$$\|Y(p)(\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))\|_2^2 = \int_{\Omega} Y(p)^2 (\partial G(\dot{\mathbb{U}}_n) - \partial G(\dot{\mathbb{U}}))^2 dx \rightarrow 0.$$

Indeed for every subsequence there exists a further subsequence such that $\dot{\mathbb{U}}_n \rightarrow \dot{\mathbb{U}}$ and we can apply the dominated convergence theorem to pass to the limit. $\square$

Lemma 1.8. The operator $\mathcal{T}_p : \mathcal{X} \rightarrow \mathcal{X}'$ is coercive uniformly in p .

Proof. For any $\xi \in \mathcal{X}$ we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &= \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{E}) (E\dot{\mathbb{U}} - \dot{\mathbb{U}}) : \left(E\dot{\mathbb{U}} - \dot{\mathbb{U}} - \frac{\Phi(\dot{\mathbb{U}})}{d} I \right) dx + \\ &+ \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \text{grad}^2 \dot{\mathbb{U}} : \text{grad}^2 \dot{\mathbb{U}} dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{\mathbb{U}} \circ \dot{\mathbb{U}} + Y(p) \partial G(\dot{\mathbb{U}}) \circ \dot{\mathbb{U}}] dx. \end{aligned}$$

Using the hypotheses (E, b) and (G, b) and using that $\dot{\mathbb{U}} : I = 0$ since $\dot{\mathbb{U}} \in \text{Dev}$ we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &\geq (\delta + \tau \gamma) \|E\dot{\mathbb{U}} - \dot{\mathbb{U}}\|_2^2 + (\alpha + \tau \epsilon) \|\text{grad } E\dot{\mathbb{U}}\|_2^2 + (\phi + \tau \beta) \|\dot{\mathbb{U}}\|_2^2 - \\ &- \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{E}) E\dot{\mathbb{U}} : \frac{\Phi(\dot{\mathbb{U}})}{d} I dx. \end{aligned}$$

Now

$$\left| \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{E}) E\dot{\mathbb{U}} : \frac{\Phi(\dot{\mathbb{U}})}{d} I dx \right| \leq \frac{\|\Phi\|_{\infty}}{d} |\mathfrak{D} + \tau \mathfrak{E}| \int_{\Omega} |E\dot{\mathbb{U}}| dx \leq C \|E\dot{\mathbb{U}}\|_2$$

for some constant $C > 0$. Moreover by the Young inequality

$$\|E\dot{\mathbb{U}} - \dot{\mathbb{U}}\|_2^2 \geq (1 - \theta) \|E\dot{\mathbb{U}}\|_2^2 + \left(1 - \frac{1}{\theta}\right) \|\dot{\mathbb{U}}\|_2^2$$

for any $\theta > 0$. Then

$$\begin{aligned} \langle \mathcal{T}_p(\xi), \xi \rangle &\geq (1 - \theta)(\delta + \tau \gamma) \|E\dot{\mathbb{U}}\|_2^2 + (\alpha + \tau \epsilon) \|\text{grad } E\dot{\mathbb{U}}\|_2^2 - C \|E\dot{\mathbb{U}}\|_2 + \\ &+ \left(1 - \frac{1}{\theta}\right) (\delta + \tau \gamma) \|\dot{\mathbb{U}}\|_2^2 + (\phi + \tau \beta) \|\dot{\mathbb{U}}\|_2^2. \end{aligned}$$

We choose $\theta < 1$ sufficiently close to 1 such there exists a constant $C_1 > 0$ with the property that

$$\left(1 - \frac{1}{\theta}\right) (\delta + \tau \gamma) \|\dot{\mathbb{U}}\|_2^2 + (\phi + \tau \beta) \|\dot{\mathbb{U}}\|_2^2 \geq C_1 \|\dot{\mathbb{U}}\|_2^2 \geq C_2 \|\dot{\mathbb{U}}\|_{1,2}^2.$$

Noticing that $C \|E\dot{\mathbb{U}}\|_2$ has subquadratic growth the thesis follows by application of Korn inequality. $\square$

Lemma 1.9. We can write $\mathcal{T}_p = \mathcal{R}_p + \mathcal{Q}$ with $\mathcal{R}_p : \mathcal{X} \rightarrow \mathcal{X}'$ and $\mathcal{Q} : \mathcal{X} \rightarrow \mathcal{X}'$ continuous and bounded operators such that

(i) $\mathcal{R}_p$ strongly monotone uniformly in p , i.e., there exists $M > 0$ such that

$$\langle \mathcal{R}_p(\xi_2) - \mathcal{R}_p(\xi_1), \xi_2 - \xi_1 \rangle \geq M \|\xi_2 - \xi_1\|_{\mathcal{X}}^2$$

for all $p \in \mathcal{Q}$ and ξ_1, ξ_2 in $\mathcal{X}$

(ii) $\mathcal{Q}$ has the property that if $\xi_n \rightarrow \xi$ in $\mathcal{X}$ then $\mathcal{Q}(\xi_n) \rightarrow \mathcal{Q}(\xi)$ in $\mathcal{X}'$ so in particular is pseudomonotone and

(iii) there exists a constant $C_{\mathcal{Q}}$ such that $\mathcal{Q}$ is Lipschitz continuous with Lipschitz constant $C_{\mathcal{Q}} \|\gamma_F\|_{\infty}$.

In particular $\mathcal{T}_p$ is pseudomonotone.

Proof. Let

$$\begin{aligned} \langle \mathcal{R}_p(\xi), \tilde{\xi} \rangle &= \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{C}) (E\dot{u} - \dot{U}) : (E\tilde{v} - \tilde{V}) \, dx + \\ &\quad + \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) \operatorname{grad}^2 \dot{u} : \operatorname{grad}^2 \tilde{v} \, dx + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B}) \dot{U} \circ \tilde{V} + Y(p) \partial G(\dot{U}) \circ \tilde{V}] \, dx \end{aligned}$$

and

$$\langle \mathcal{Q}(\xi), \tilde{\xi} \rangle = - \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{C}) \frac{\Phi(\dot{U})}{d} I : (E\tilde{v} - \tilde{V}) \, dx.$$

First we prove that $\mathcal{R}_p$ is strongly monotone uniformly in p . For all ξ_1 and ξ_2 in $\mathcal{X}$ we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle &= \\ &= \int_{\Omega} (\mathfrak{D} + \tau \mathfrak{C}) (E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)) : (E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)) \, dx + \\ &\quad + \int_{\Omega} (\mathfrak{E} + \tau \mathfrak{A}) (\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1) : (\operatorname{grad}^2 \dot{u}_2 - \operatorname{grad}^2 \dot{u}_1) \, dx + \\ &\quad + \int_{\Omega} [(\mathfrak{F} + \tau \mathfrak{B})(\dot{U}_2 - \dot{U}_1) \circ (\dot{U}_2 - \dot{U}_1) + Y(p)(\partial G(\dot{U}_2) - \partial G(\dot{U}_1)) \circ (\dot{U}_2 - \dot{U}_1)] \, dx \geq \\ &\geq (\delta + \tau \gamma) \|E(\dot{u}_2 - \dot{u}_1) - (\dot{U}_2 - \dot{U}_1)\|_2^2 + (\phi + \tau \beta) \|\dot{U}_2 - \dot{U}_1\|_2^2. \end{aligned}$$

We have used that since G is convex then ∂G is monotone. Reasoning as in the proof of Lemma 1.8 we have

$$\begin{aligned} \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle &\geq (1 - \theta)(\delta + \tau \gamma) \|E(\dot{u}_2 - \dot{u}_1)\|_2^2 + (\alpha + \tau \epsilon) \|\operatorname{grad} E(\dot{u}_2 - \dot{u}_1)\|_2^2 + \\ &\quad + \left(1 - \frac{1}{\theta}\right) (\delta + \tau \gamma) \|\dot{U}_2 - \dot{U}_1\|_2^2 + \\ &\quad + (\phi + \tau \beta) \|\dot{U}_2 - \dot{U}_1\|_2^2. \end{aligned}$$

Choosing $0 < \theta < 1$ sufficiently close to 1 and using Korn inequality we conclude. Next we prove the properties of $\mathcal{Q}$. Let $\xi_n \rightarrow \xi$ in $\mathcal{X}$. Then $\dot{U}_n \rightarrow \dot{U}$ in H^1 and by Proposition B.1 we have $\dot{U}_n \rightarrow \dot{U}$ in L^2 . Since Φ is Lipschitz we have $\Phi(\dot{U}_n) \rightarrow \Phi(\dot{U})$ in L^2 . This establish the first property which in turns implies that $\mathcal{Q}$ is pseudomonotone. Next we observe that

$$|\langle \mathcal{Q}(\xi_2) - \mathcal{Q}(\xi_1), \tilde{\xi} \rangle| \leq \int_{\Omega} |(\mathfrak{D} + \tau \mathfrak{C}) \operatorname{Lip}(\Phi) \dot{U}_2 - \dot{U}_1| (|E\tilde{v}| + |\tilde{V}|) \, dx \leq C \|\gamma_F\|_{\infty} \|\xi_2 - \xi_1\|_{\mathcal{X}} \|\tilde{\xi}\|_{\mathcal{X}}.$$

This establish also the second property. Now by Proposition B.18 and Proposition B.19 we have that $\mathcal{T}_p$ is pseudomonotone. $\square$

Now applying Theorem B.15 for each $p \in \mathcal{Q}$ there exists $\xi \in \mathcal{X}$ such that

$$\langle \mathcal{T}_p(\xi), \tilde{\zeta} \rangle = \langle l, \tilde{\mathcal{J}}(\tilde{\zeta}) \rangle \quad \text{for all } \tilde{\zeta} \in \mathcal{X}.$$

Moreover ξ is unique if $\|\gamma_F\|_\infty$ is sufficiently small. Indeed if ξ_1 and ξ_2 are two solutions we have

$$\begin{aligned} M\|\xi_2 - \xi_1\|_{\mathcal{X}}^2 &\leq \langle \mathcal{R}_p(\xi_2) - \mathcal{R}_p(\xi_1), \xi_2 - \xi_1 \rangle \leq \\ &= \langle \mathcal{T}_p(\xi_2) - \mathcal{T}_p(\xi_1), \xi_2 - \xi_1 \rangle - \langle \mathcal{Q}(\xi_2) - \mathcal{Q}(\xi_1), \xi_2 - \xi_1 \rangle = \\ &= -\langle \mathcal{Q}(\xi_2) - \mathcal{Q}(\xi_1), \xi_2 - \xi_1 \rangle \leq C_Q \|\gamma_F\|_\infty \|\xi_2 - \xi_1\|_{\mathcal{X}}^2. \end{aligned}$$

We call $\Xi: \mathcal{Q} \rightarrow \mathcal{X}$ the map that assign to $p \in \mathcal{Q}$ the unique ξ . Now to conclude the existence of a solution to Problem 1.5 we need to solve in $p \in \mathcal{Q}$ the equation

$$\langle \mathcal{A}(p, \mathcal{J}(\Xi(p))), (0, 0, \tilde{\mu}) \rangle + \langle Bp, (0, 0, \tilde{\mu}) \rangle = \langle l, (0, 0, \tilde{\mu}) \rangle \quad \text{for all } \tilde{\mu} \in L^2(\Omega)$$

that is

$$\int_{\Omega} \text{tr} \left[\left(\mathfrak{D} + \tau \mathfrak{C} \right) \left(\mathbb{E} \dot{u} - \dot{u} - \frac{\dot{\lambda}}{d} I \right) \right] \frac{\tilde{\mu}}{d} dx - \int_{\Omega} p \tilde{\mu} dx = \int_{\Omega} \text{tr} \left[\mathfrak{C} \left(\mathbb{E} u - u - \frac{\lambda}{d} I \right) \right] \frac{\tilde{\mu}}{d} dx$$

for all $\tilde{\mu} \in L^2(\Omega)$, where $\dot{y} = (\dot{u}, \dot{u}, \dot{\lambda}) = \mathcal{J}(\Xi(p))$. This is equivalent to

$$p = \frac{1}{d} \text{tr} \left[\left(\mathfrak{D} + \tau \mathfrak{C} \right) \left(\mathbb{E} \dot{u} - \dot{u} - \frac{\dot{\lambda}}{d} I \right) \right] - \frac{1}{d} \text{tr} \left[\mathfrak{C} \left(\mathbb{E} u - u - \frac{\lambda}{d} I \right) \right] = \mathcal{S}(p)$$

where we have introduced the operator $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$. We want to prove the existence of a fixed point of $\mathcal{S}$ using the Schauder fixed point theorem (Theorem B.11). To prove that $\mathcal{S}$ satisfy the required hypotheses we need to establish some regularity results regarding Ξ and $\mathcal{J}$. We first need the following lemma.

Lemma 1.10. The operator $\mathcal{A}$ is Lipschitz continuous in p uniformly in $\dot{y}$.

Proof. By (Y, a) and (G, c) we have

$$\begin{aligned} |\langle \mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y}), \tilde{z} \rangle| &\leq \int_{\Omega} |Y(p) - Y(q)| |\partial G(\dot{U})| |\tilde{V}| dx \leq \\ &\leq \text{Lip}(Y) \|\partial G\|_\infty \int_{\Omega} |p - q| |\tilde{V}| dx \leq L \|p - q\|_2 \|\tilde{V}\|_{1,2} \end{aligned}$$

for some constant $L > 0$. Then we have

$$\|\mathcal{A}(p, \dot{y}) - \mathcal{A}(q, \dot{y})\|_{\tilde{z}'} \leq L \|p - q\|_{\mathcal{Q}}$$

This concludes the proof. $\square$

Now we can prove the continuity of Ξ .

Lemma 1.11. The operator Ξ is continuous and has bounded range.

Proof. First we prove that Ξ has bounded range. We assume by contradiction that there exists $p_n \in \mathcal{Q}$ such that $\|\Xi(p_n)\|_{\mathcal{X}} \geq n$. Let $\xi_n = \Xi(p_n)$. Now

$$\frac{\langle \mathcal{T}_{p_n}(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \leq \|\langle l, \tilde{\mathcal{J}}(\cdot) \rangle\|_{\mathcal{X}'}$$

But by Lemma 1.8

$$\frac{\langle \mathcal{T}_p(\xi_n), \xi_n \rangle}{\|\xi_n\|_{\mathcal{X}}} \rightarrow \infty$$

and this is a contradiction. Now we prove that Ξ is continuous. Let $p_n \rightarrow p$ in $\mathcal{Q}$ and let $\xi_n = \Xi(p_n)$, $\xi = \Xi(p)$. By the first part of the proof we know that ξ_n are bounded in $\mathcal{X}$. Then for every subsequence of ξ_n there exists a further subsequence that converges weakly in $\mathcal{X}$. If we prove that each of this weakly convergent subsequences converges strongly to ξ we have done. So taking a subsequence we assume that $\xi_n \rightharpoonup \zeta$ in $\mathcal{X}$ for some $\zeta \in \mathcal{X}$. We have to prove that $\xi_n \rightarrow \xi$ in $\mathcal{X}$. Now using strong monotonicity of $\mathcal{R}_p$ uniformly in p and that $\mathcal{T}_p(\xi) = \mathcal{T}_{p_n}(\xi_n)$ we have

$$\begin{aligned} M\|\xi_n - \xi\|_{\mathcal{X}}^2 &\leq \langle \mathcal{R}_p(\xi_n) - \mathcal{R}_p(\xi), \xi_n - \xi \rangle = \\ &= \langle \mathcal{T}_p(\xi_n) - \mathcal{T}_p(\xi), \xi_n - \xi \rangle - \langle \mathcal{Q}(\xi_n) - \mathcal{Q}(\xi), \xi_n - \xi \rangle \leq \\ &\leq \langle \mathcal{T}_p(\xi_n) - \mathcal{T}_{p_n}(\xi_n), \xi_n - \xi_n \rangle - \langle \mathcal{Q}(\xi_n) - \mathcal{Q}(\xi), \xi_n - \xi \rangle. \end{aligned}$$

But since $\mathcal{A}$ is Lipschitz in p uniformly in $\dot{y}$ we have

$$\langle \mathcal{T}_p(\xi_n) - \mathcal{T}_{p_n}(\xi_n), \xi_n - \xi_n \rangle \leq L\|p_n - p\|_{\mathcal{Q}}\|\xi_n - \xi\|_{\mathcal{X}} \rightarrow 0$$

where we use that ξ_n are bounded since they converge weakly. Moreover by Proposition 1.9 we know that $\mathcal{Q}(\xi_n) \rightarrow \mathcal{Q}(\xi)$ in $\mathcal{X}'$. Then by a standard result we have $\langle \mathcal{Q}(\xi_n), \xi_n \rangle$. It follows that

$$\langle \mathcal{Q}(\xi_n) - \mathcal{Q}(\xi), \xi_n - \xi \rangle \rightarrow 0.$$

We have proved that $\xi_n \rightarrow \xi$ in $\mathcal{X}$. □

The following lemma establish the regularity of $\mathcal{J}$.

Lemma 1.12. The map $\mathcal{J}: \mathcal{X} \rightarrow \mathcal{Z}$ is continuous and bounded.

Proof. We recall that if $\xi = (\dot{u}, \dot{U}) \in \mathcal{X}$ then $\mathcal{J}(\xi) = (\dot{u}, \dot{U}, \Phi(\dot{U}))$. It is enough to prove that $\Phi: H^1 \rightarrow L^2$ is continuous and bounded. Boundedness follows immediately from the estimates of Lemma 1.6. Continuity follows from the fact that Φ is Lipschitz as a function from DevSym to $\mathbb{R}$. □

Now we are ready to prove the desired properties of $\mathcal{S}$.

Lemma 1.13. The map $\mathcal{S}: \mathcal{Q} \rightarrow \mathcal{Q}$ is continuous and has a bounded and compact range.

Proof. Notice that the map

$$\mathcal{J}(\dot{y}) = \frac{1}{d} \text{tr} \left[(\mathfrak{D} + \tau \mathfrak{C}) \left(E\dot{u} - \dot{U} - \frac{1}{d} \frac{\dot{\lambda}}{\dot{d}} I \right) \right] - \text{tr} \left[\mathfrak{C} \left(Eu - U - \frac{\lambda}{d} I \right) \right]$$

is continuous, compact and bounded from $\mathcal{Z}$ to $\mathcal{Q}$. The compactness follows since H^1 is compactly embedded in L^2 by Rellich–Kondrachov theorem. Then by Lemma 1.11 and Lemma 1.11 we have that $\mathcal{S}$ is continuous and has a compact and bounded range. Indeed the continuity follows since $\mathcal{S}$ is a composition of continuous functions. The boundedness of the range is a trivial consequence of the boundedness of the range of Ξ and the fact that both $\mathcal{J}$ and $\mathcal{J}$ are bounded. Moreover the compactness of the range is established as follows. Let $p_n \in \text{range}(\mathcal{S})$ and let $q_n \in \mathcal{Q}$ such that $p_n = \mathcal{S}(q_n)$. We know that $\dot{y}_n = \mathcal{J}(\Xi(q_n))$ is bounded in $\mathcal{Z}$. By compactness of $\mathcal{J}$ then $p_n = \mathcal{J}(\dot{y}_n)$ admits a convergent subsequence. □

Now we are in position to apply Schauder fixed point theorem and we get the existence of a fixed point of $\mathcal{S}$. We have obtained the desired result.

Proposition 1.14. If $\|\gamma_F\|_\infty$ is sufficiently small then Problem 1.5 admits at least a solution.

Now we need to pass to the limit $N \rightarrow \infty$. We will try to get the a priori estimates needed to apply Aubin–Lions lemma. For simplicity we start with the case $F = 0$. Our problem is of the form

$$\begin{cases} Ay + E\dot{y} + D(p, \dot{y}) + Bp = f \\ B^T \dot{y} = 0. \end{cases}$$

We have obtained

$$\begin{cases} Ay_n + E\dot{y}_n + D(p_n, \dot{y}_n) + Bp_n = f_n \\ B^T \dot{y}_n = 0 \end{cases}$$

where $y_n = y_{n-1} + \tau y_n$. We test the first equation with $\dot{y}_n$ and we notice that $\langle D(p_n, \dot{y}_n), \dot{y}_n \rangle \geq 0$ and $\langle Bp_n, \dot{y}_n \rangle = 0$. Then

$$\langle \tau A \dot{y}_n, \dot{y}_n \rangle + \langle E \dot{y}_n, \dot{y}_n \rangle \leq \langle f_n, \dot{y}_n \rangle - \langle Ay_{n-1}, \dot{y}_n \rangle.$$

Since A and E are coercive we have

$$\tau \alpha \|\dot{y}_n\|_Z + \epsilon \|\dot{y}_n\|_Z \leq \|f_n\|_{Z'} + M \|y_{n-1}\|_Z.$$

By Gronwall inequality we have that y_n is bounded in Z uniformly in n and τ . Then also $\dot{y}_n$ is bounded in Z uniformly in n and τ . Moreover since p_n is given by the expression above it is bounded in Q uniformly in n and τ .

1.2.5 Local existence and uniqueness theorem

Lemma 1.15. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $\phi: \mathbb{R}^m \rightarrow \mathbb{R}^l$ be Lipschitz continuous and bounded. Let $f_n \rightarrow f$ in $L^p(\Omega, \mathbb{R}^m)$ for some $p \in [1, \infty)$. Then $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$ for all $q \in [p, \infty)$.

Proof. Let $A_n = \{x \in \Omega \mid |f_n(x) - f(x)| < 1\}$. Since convergence in L^p implies convergence in measure then $|\Omega \setminus A_n| \rightarrow 0$. Now let $q \in [p, \infty)$. Then if we call $L > 0$ the Lipschitz constant of ϕ and $C = \sup_{\mathbb{R}^m} |\phi|$ we have

$$\begin{aligned} \|\phi(f_n) - \phi(f)\|_q^q &= \int_{A_n} |\phi(f_n) - \phi(f)|^q dx + \int_{\Omega \setminus A_n} |\phi(f_n) - \phi(f)|^q dx \leq \\ &\leq \int_{A_n} L |f_n - f|^q dx + 2C |\Omega \setminus A_n|. \end{aligned}$$

But on A_n we have $|f_n - f|^q \leq |f_n - f|^p$ so that

$$\|\phi(f_n) - \phi(f)\|_q^q \leq \int_{A_n} L |f_n - f|^p dx + 2C |\Omega \setminus A_n| \leq L \|f_n - f\|_p^p + 2C |\Omega \setminus A_n|.$$

This proves that $\phi(f_n) \rightarrow \phi(f)$ in $L^q(\Omega, \mathbb{R}^l)$. $\square$

Now we call $\mathcal{X} = \mathcal{Z} \times \mathcal{Q}$ and $\tilde{\mathcal{X}} = \tilde{\mathcal{Z}} \times \tilde{\mathcal{Q}}$. We will write $\xi = (\dot{y}, p) \in \mathcal{X}$, $\zeta = (z, q) \in \mathcal{X}$ and $\tilde{\zeta} = (\tilde{z}, \tilde{q}) \in \tilde{\mathcal{X}}$. Now we can view $\mathcal{F}: \mathcal{X} \times \mathcal{Z} \times [0, T] \rightarrow \tilde{\mathcal{X}}$.

Proposition 1.16. The map $\mathcal{F}$ is of class C^1 .

Proof. We need to check the thesis only for the nonlinear terms (1.2.1). We first study the Fréchet differentiability of the second term in (1.2.1) with respect to λ . By the dominated convergence theorem we have

$$\int_{\Omega} \left(\frac{F(\lambda + \tau\mu, \dot{U}) - F(\lambda, \dot{U})}{\tau} - \partial_{\lambda} F(\lambda, \dot{U})\mu \right) \tilde{q} dx \rightarrow 0$$

for $\tau \rightarrow 0$. Indeed

$$\left| \frac{F(\lambda + \tau\mu, \dot{U}) - F(\lambda, \dot{U})}{\tau} - \partial_{\lambda} F(\lambda, \dot{U})\mu \right| |\tilde{q}| \leq 2\text{Lip}(F)|\mu||\tilde{q}| \in L^1.$$

So we have proved the Gateaux differentiability. Now we prove that the Gateaux derivative is continuous. Let $\dot{y}_n \rightarrow \dot{y}$ in $\mathcal{Z}$, $p_n \rightarrow p$ in $\mathcal{Q}$ and $y_n \rightarrow y$ in $\mathcal{Z}$. Then

$$\left| \int_{\Omega} (\partial_{\lambda} F(\lambda_n, \dot{U}_n) - \partial_{\lambda} F(\lambda, \dot{U})) \mu \tilde{q} dx \right| \leq \int_{\Omega} \text{Lip}(\partial_{\lambda} F)$$

We claim that the Gateaux derivative of $\mathcal{F}$ with respect to ξ is

$$\begin{aligned} \langle D_{\xi} \mathcal{F}(\xi, y, t) \zeta, \tilde{\zeta} \rangle &= \int_{\Omega} \mathfrak{D} \left(E v - V - \frac{\mu}{d} I \right) : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &+ \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) \mathbb{V} \circ \tilde{\mathbb{V}} dx + \int_{\Omega} Y'(p) q \partial G(\dot{U}) \circ \tilde{\mathbb{V}} dx - \\ &- \int_{\Omega} q \tilde{\mu} dx + \int_{\Omega} (\partial_{\dot{U}} F(\lambda, \dot{U}) : V - \mu) \tilde{q} dx. \end{aligned}$$

We need only to check the claim for the terms (1.2.1). We start with the Gateaux derivative with respect to p of the first term. We notice that by Hölder inequality and by the continuous embedding of H^1 in L^6

$$\begin{aligned} \int_{\Omega} \left| \frac{Y(p + \epsilon q) - Y(p)}{\epsilon} - Y'(p) \right| |G'(\dot{\xi})| |\tilde{\eta}| dx &\leq \left\| \frac{Y(p + \epsilon q) - Y(p)}{\epsilon} - Y'(p) \right\|_{3/2} \|G'(\dot{\xi})\|_6 \|\tilde{\eta}\|_6 \lesssim \\ &\lesssim \left(\int_{\Omega} \left| \frac{Y(p + \epsilon q) - Y(p)}{\epsilon} - Y'(p) \right|^{3/2} dx \right)^{2/3} (1 + \|\dot{\xi}\|_{1,2}) \|\tilde{\eta}\|_{1,2}. \end{aligned}$$

Since

$$\lim_{\epsilon \rightarrow 0} \left(\int_{\Omega} \left| \frac{Y(p + \epsilon q) - Y(p)}{\epsilon} - Y'(p) \right|^{3/2} dx \right)^{2/3} = 0$$

by dominated convergence we conclude the check by applying Lemma B.7. Now we turn to the Gateaux derivative of the first term in (1.2.1) with respect to $\dot{\xi}$. We have

$$\begin{aligned} \int_{\Omega} Y(p) \left| \frac{G'(\dot{\xi} + \epsilon \eta) - G'(\dot{\xi})}{\epsilon} - G''(\dot{\xi})\eta \right| |\tilde{\eta}| dx &\leq C_Y \left\| \frac{G'(\dot{\xi} + \epsilon \eta) - G'(\dot{\xi})}{\epsilon} - G''(\dot{\xi})\eta \right\|_{6/5} \|\tilde{\eta}\|_6 \lesssim \\ &\lesssim \left(\int_{\Omega} \left| \frac{G'(\dot{\xi} + \epsilon \eta) - G'(\dot{\xi})}{\epsilon} - G''(\dot{\xi})\eta \right|^{6/5} dx \right)^{5/6} \|\tilde{\eta}\|_{1,2}. \end{aligned}$$

By hypotheses (G, c) we know that

$$\left| \frac{G'(\dot{\xi} + \epsilon \eta) - G'(\dot{\xi})}{\epsilon} - G''(\dot{\xi})\eta \right| \leq 2L_G |\eta| \in L^{6/5}(\Omega).$$

Then by dominated convergence

$$\lim_{\epsilon \rightarrow 0} \left(\int_{\Omega} \left| \frac{G'(\dot{\xi} + \epsilon \eta) - G'(\dot{\xi})}{\epsilon} - G''(\dot{\xi})\eta \right|^{6/5} dx \right)^{5/6} = 0$$

and we conclude the check by Lemma B.7. Eventually we check the Gateaux derivative with respect to $\dot{\xi}$ of the third term in (1.2.1). We have

$$\int_{\Omega} \left| \frac{F(\dot{\xi} + \epsilon \eta) - F(\dot{\xi})}{\epsilon} - F'(\dot{\xi}) : \eta \right| |\tilde{q}| dx \leq \left(\int_{\Omega} \left| \frac{F(\dot{\xi} + \epsilon \eta) - F(\dot{\xi})}{\epsilon} - F'(\dot{\xi}) : \eta \right|^2 dx \right)^{1/2} \|\tilde{q}\|_2.$$

Using hypotheses (F, c) we have

$$\left| \frac{F(\dot{\xi} + \epsilon \eta) - F(\dot{\xi})}{\epsilon} - F'(\dot{\xi}) : \eta \right| \leq 2L_F |\eta| \in L^2(\Omega).$$

Then by dominated convergence and the application of Lemma B.7 we complete also this check.

Step 4. To check that $\mathcal{F}$ is also Fréchet differentiable with respect to α it is enough to prove the continuity of the Gateaux derivative. We need to prove it only for the terms

$$\int_{\Omega} Y(p)G''(\dot{\xi})\eta : \tilde{\eta} dx, \quad \int_{\Omega} Y'(p)qG'(\dot{\xi}) : \tilde{\eta} dx \quad \text{and} \quad \int_{\Omega} (F'(\dot{\xi}) : \eta - \mu)\tilde{q} dx.$$

Let $p_n \rightarrow p$ in $L^2(\Omega)$ and $\dot{\xi}_n \rightarrow \dot{\xi}$ in $H^1(\Omega, \text{DevSym})$. We study the continuity of the first term. Using the Hölder inequality we have

$$\begin{aligned} \int_{\Omega} |Y(p_n)G''(\dot{\xi}_n) - Y(p)G''(\dot{\xi})| |\eta| |\tilde{\eta}| dx &\leq \|Y(p_n)G''(\dot{\xi}_n) - Y(p)G''(\dot{\xi})\|_{3/2} \|\eta\|_6 \|\tilde{\eta}\|_6 \lesssim \\ &\lesssim \|Y(p_n)G''(\dot{\xi}_n) - Y(p)G''(\dot{\xi})\|_{3/2} \|\eta\|_{1,2} \|\tilde{\eta}\|_{1,2}. \end{aligned}$$

Now

$$\begin{aligned} \|Y(p_n)G''(\dot{\xi}_n) - Y(p)G''(\dot{\xi})\|_{3/2} &\leq \\ &\leq \|(Y(p_n) - Y(p))G''(\dot{\xi})\|_{3/2} + \|Y(p_n)(G''(\dot{\xi}_n) - G''(\dot{\xi}))\|_{3/2} \leq \\ &\leq L_Y L_G \|p_n - p\|_{3/2} + C_Y \text{Lip}(G'') \|\dot{\xi}_n - \dot{\xi}\|_{3/2} \lesssim \\ &\lesssim \|p_n - p\|_2 + \|\dot{\xi}_n - \dot{\xi}\|_{1,2} \rightarrow 0. \end{aligned}$$

Using Lemma B.8 we have the continuity of the first term. Next we study the second term. We have

$$\begin{aligned} \int_{\Omega} |Y'(p_n)G'(\dot{\xi}_n) - Y'(p)G'(\dot{\xi})| |q| |\tilde{\eta}| dx &\leq \|Y'(p_n)G'(\dot{\xi}_n) - Y'(p)G'(\dot{\xi})\|_3 \|q\|_2 \|\tilde{\eta}\|_6 \lesssim \\ &\lesssim \|Y'(p_n)G'(\dot{\xi}_n) - Y'(p)G'(\dot{\xi})\|_3 \|q\|_2 \|\tilde{\eta}\|_{1,2}. \end{aligned}$$

Now

$$\begin{aligned} \|Y'(p_n)G'(\dot{\xi}_n) - Y'(p)G'(\dot{\xi})\|_3 &\leq \|(Y'(p_n) - Y'(p))G'(\dot{\xi})\|_3 + \|Y'(p_n)(G'(\dot{\xi}_n) - G'(\dot{\xi}))\|_3 \leq \\ &\leq C_G \|(1 + |\dot{\xi}|)(Y'(p_n) - Y'(p))\|_3 + L_Y L_G \|\dot{\xi}_n - \dot{\xi}\|_3 \leq \\ &\lesssim (1 + \|\dot{\xi}\|_6) \|Y'(p_n) - Y'(p)\|_6 + \|\dot{\xi}_n - \dot{\xi}\|_{1,2} \rightarrow 0 \end{aligned}$$

where the last limit holds because by Lemma 1.15 we have $Y'(p_n) \rightarrow Y'(p)$ in $L^6(\Omega)$. By Lemma B.8 we get that also the second term is continuous. We now turn to the check of the continuity of the last term. We have

$$\begin{aligned} \int_{\Omega} |F'(\dot{\xi}) - F'(\dot{\xi}_n)| |\eta| |\tilde{q}| dx &\leq \|F'(\dot{\xi}) - F'(\dot{\xi}_n)\|_3 \|\eta\|_6 \|\tilde{q}\|_2 \leq \\ &\leq \text{Lip}(F') \|\dot{\xi} - \dot{\xi}_n\|_3 \|\eta\|_6 \|\tilde{q}\|_2 \lesssim \|\dot{\xi} - \dot{\xi}_n\|_{1,2} \|\eta\|_{1,2} \|\tilde{q}\|_2. \end{aligned}$$

Applying Lemma B.8 we get the continuity of the last term. $\square$

Proposition 1.17. For any $\dot{y} \in \mathcal{Z}$, $p \in \mathcal{Q}$ and $y \in \mathcal{Z}$ with $\|\dot{U}\|_{1,2}$ sufficiently small we have $D_{\xi} \mathcal{F}(\xi, y, t) \in \text{Iso}(\mathcal{X}, \tilde{\mathcal{X}}')$.

Proof. We define the bilinear form $a: \mathcal{Z} \times \tilde{\mathcal{Z}} \rightarrow \mathbb{R}$ by

$$\begin{aligned} a(z, \tilde{z}) &= \int_{\Omega} \mathfrak{D} \left(E v - V - \frac{\mu}{d} I \right) : \left(E \tilde{v} - \tilde{V} - \frac{\tilde{\mu}}{d} I \right) dx + \\ &+ \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 \tilde{v} dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) \nabla \circ \tilde{\nabla} dx. \end{aligned}$$

We define also the bilinear forms $b: \mathcal{Q} \times \tilde{\mathcal{Z}} \rightarrow \mathbb{R}$ and $c: \tilde{\mathcal{Q}} \times \mathcal{Z} \rightarrow \mathbb{R}$ by

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) \circ \tilde{\nabla} - \tilde{\mu}) q dx$$

and

$$c(\tilde{q}, z) = \int_{\Omega} (\partial_{\dot{U}} F(\lambda, \dot{U}) : V - \mu) \tilde{q} dx = 0.$$

To prove that $D_{\xi} \mathcal{F}(\xi, y, t) \in \text{Iso}(\mathcal{X}, \tilde{\mathcal{B}}')$ we can prove that the mixed elliptic problem

$$\begin{aligned} a(z, \tilde{z}) + b(p, \tilde{z}) &= \langle l, \tilde{z} \rangle \quad \text{for all } \tilde{v} \in \tilde{\mathcal{V}} \\ c(\tilde{q}, z) &= \langle m, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{\mathcal{Q}} \end{aligned}$$

is well posed for all $l \in \tilde{\mathcal{V}}'$ and $m \in \tilde{\mathcal{Q}}'$. To this aim it is sufficient to check the hypotheses of Theorem B.6. We first check the hypotheses on b . We have

$$\tilde{\mathcal{N}} = \{ \tilde{z} \in \tilde{\mathcal{Z}} \mid b(p, \tilde{z}) = 0 \text{ for all } p \in \mathcal{Q} \} = \{ \tilde{z} \in \tilde{\mathcal{Z}} \mid \tilde{\mu} = Y'(p) \partial G(\dot{U}) : \tilde{\nabla} \}.$$

Notice that $|Y'(p) \partial G(\dot{U}) : \tilde{\nabla}| \leq \text{Lip}(Y) \|\partial G\|_{\infty} \|\nabla\| \in L^2$. Now

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) \circ \tilde{\nabla} - \tilde{\mu}) q dx}{\|q\|_Q} = \|Y'(p) \partial G(\dot{U}) \circ \tilde{\nabla} - \tilde{\mu}\|_2.$$

On the other hand since $\tilde{z}_0 = (\tilde{v}_0, \tilde{V}_0, \tilde{\mu}_0) \in \tilde{\mathcal{N}}$ if and only if $\tilde{\mu}_0 = Y'(p) \partial G(\dot{U}) : \tilde{\nabla}_0$ we have

$$\begin{aligned} \|\tilde{z}\|_{\tilde{\mathcal{Z}}/\tilde{\mathcal{N}}} &= \inf_{\tilde{z}_0 \in \tilde{\mathcal{N}}} [\|\tilde{v} + \tilde{v}_0\|_{1,2} + \|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{\nabla}_0\|_2] = \\ &= \inf_{\tilde{V}_0 \in H^1} [\|\tilde{V} + \tilde{V}_0\|_{1,2} + \|\tilde{\mu} + Y'(p) \partial G(\dot{U}) : \tilde{\nabla}_0\|_2] \leq \\ &\leq \|\tilde{\mu} - Y'(p) \partial G(\dot{U}) : \tilde{\nabla}\|_2 \end{aligned}$$

where the last inequality holds since we can choose $\tilde{\nabla}_0 = -\tilde{\nabla}$. Then we have

$$\sup_{q \in \mathcal{Q} \setminus \{0\}} \frac{\int_{\Omega} (Y'(p) \partial G(\dot{U}) \circ \tilde{\nabla} - \tilde{\mu}) q dx}{\|q\|_Q} \geq \|\tilde{z}\|_{\tilde{\mathcal{Z}}/\tilde{\mathcal{N}}}.$$

Moreover $q \in \mathcal{Q}$ such that

$$b(p, \tilde{z}) = \int_{\Omega} (Y'(p) \partial G(\dot{U}) \circ \tilde{V} - \tilde{\mu}) q \, dx = 0 \quad \text{for all } \tilde{z} \in \tilde{\mathcal{Z}}.$$

Choosing $\tilde{V} = 0$ and $\tilde{\mu} \in L^2(\Omega)$ arbitrary we have $q = 0$. This completes the check of the hypotheses on b . The check of the hypotheses on c is similar. In particular we have

$$\mathcal{N} = \{z \in \mathcal{Z} \mid c(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{\mathcal{Q}}\} = \{z \in \mathcal{Z} \mid \mu = \partial_{\dot{U}} F(\lambda, \dot{U}) : V\}$$

and

$$\sup_{\tilde{q} \in \tilde{\mathcal{Q}} \setminus \{0\}} \frac{\int_{\Omega} (\partial_{\dot{U}} F(\lambda, \dot{U}) : V - \mu) \tilde{q} \, dx}{\|\tilde{q}\|_{\tilde{\mathcal{Q}}}} \geq \|z\|_{\mathcal{Z}/\mathcal{N}}$$

and if $\tilde{q} \in \tilde{\mathcal{Q}}$ satisfies

$$\int_{\Omega} (\partial_{\dot{U}} F(\lambda, \dot{U}) : V - \mu) \tilde{q} \, dx = 0 \quad \text{for all } z \in \mathcal{Z}$$

then $\tilde{q} = 0$. Now we check the hypotheses on a . Let $z \in \mathcal{Z} = \tilde{\mathcal{Z}}$. Then

$$\begin{aligned} a(z, z) &\geq \delta \left\| Ev - V - \frac{\mu}{d} I \right\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + c_Y \gamma_G(\dot{U}) \|\mathbb{V}\|_2^2 \geq \\ &\geq (1 - \theta) \delta \|Ev\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + \left(1 - \frac{1}{\theta}\right) \delta \left\| V + \frac{\mu}{d} I \right\|_2^2 + c_Y \gamma_G(\dot{U}) \|\mathbb{V}\|_2^2 = \\ &= (1 - \theta) \delta \|Ev\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + \left(1 - \frac{1}{\theta}\right) \delta \|V\|_2^2 + \left(1 - \frac{1}{\theta}\right) \frac{\delta}{d} \|\mu\|_2^2 + c_Y \gamma_G(\dot{U}) \|\mathbb{V}\|_2^2 \end{aligned}$$

for any $\theta > 0$. We have used Young inequality and the fact that $V : I = 0$ since $V \in \text{Dev}$. Now if $\dot{U} = 0$ by (F, e) and (G, f) we have

$$\mathcal{N} = \tilde{\mathcal{N}} = \{z \in \mathcal{Z} \mid \mu = 0\}.$$

This implies that if $z \in \mathcal{N}$ then

$$a(z, z) \geq (1 - \theta) \delta \|Ev\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + \left(1 - \frac{1}{\theta}\right) \delta \|V\|_2^2 + c_Y \gamma_G(0) \|\mathbb{V}\|_2^2.$$

choosing $\theta < 1$ sufficiently close to 1 such that

$$c_Y \gamma_G(0) + \left(1 - \frac{1}{\theta}\right) \delta > 0$$

by applying the second order Korn inequality we get that a is coercive in $\mathcal{N}$. Then also the hypotheses on a are verified. This implies that if $\dot{U} = 0$ then $D_{\xi} \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{X}, \tilde{\mathcal{X}}')$. But since $\mathcal{F}$ is of class C^1 then for $\|\dot{U}\|_{1,2}$ sufficiently small we still have $D_{\xi} \mathcal{F}(\dot{y}, p, y, t) \in \text{Iso}(\mathcal{X}, \tilde{\mathcal{X}}')$. $\square$

Remark. Let $z \in \mathcal{N}$ and $\tilde{z} \in \tilde{\mathcal{N}}$ such that $\tilde{v} = v$ and $\tilde{V} = V$. Then

$$\begin{aligned} a(z, \tilde{z}) &= \int_{\Omega} \mathfrak{D} \left(Ev - V - \frac{\partial_{\dot{U}} F(\lambda, \dot{U}) : V}{d} I \right) : \left(Ev - V - \frac{Y'(p) \partial G(\dot{U}) \circ \mathbb{V}}{d} I \right) dx + \\ &+ \int_{\Omega} \mathfrak{E} \text{grad}^2 v : \text{grad}^2 v \, dx + \int_{\Omega} Y(p) \partial^2 G(\dot{U}) \mathbb{V} \circ \tilde{\mathbb{V}} \, dx \geq \\ &\geq \delta (1 - \theta) \|Ev\|_2^2 + \delta \left(1 - \frac{1}{\theta}\right) \|V\|_2^2 - C \|Ev\|_2 \|\mathbb{V}\|_2 + \epsilon \|\text{grad } Ev\|_2^2 + c_Y \gamma_G(\dot{U}) \|\mathbb{V}\|_2^2 \end{aligned}$$

for some $C > 0$ and all $\theta > 0$. Then

$$\alpha(z, \tilde{z}) \geq [\delta(1 - \theta) - C\eta] \|Ev\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + \delta \left(1 - \frac{1}{\theta}\right) \|V\|_2^2 + \left(c_Y \gamma_G(\dot{U}) - \frac{C}{\eta}\right) \|\mathbb{V}\|_2^2.$$

This guarantees the hypotheses of Theorem B.2 on α only if $c_Y \gamma_G(\dot{U})$ is sufficiently large. This suggests that $D_\alpha \mathcal{F}$ tends to become singular during the time evolution. This inevitably leads to the loss of uniqueness of the solution. We hope that the bifurcation is associated with the formation of shear bands according to the mechanism we now describe. When the body is subjected to an external load, it begins to flow uniformly plastically, and $\dot{U}$ grows uniformly across the entire domain. When $\dot{U}$ becomes too large, the coercivity of α is lost, and the bifurcation occurs. In the new stable branch, $\dot{U}$ decreases near to zero almost everywhere except at the shear band. This restores the coercivity of α because where $\dot{U}$ is small, the plastic dissipation anchors V , preventing it from growing indefinitely, while in the shear band region, V clings to adjacent through the nonlocal dissipation.

Remark. Consider the particular case

$$\mathcal{N} = \tilde{\mathcal{N}} = \{(v, V, \mu) \in \mathcal{Z} \mid \mu = H \circ \mathbb{V}\}$$

for some $H \in \text{DevPla}$. This corresponds to the associative case in the classical Drucker–Prager model. Then given $z \in \mathcal{N}$ we have

$$\alpha(z, z) \geq (1 - \theta)\delta \|Ev\|_2^2 + \epsilon \|\text{grad } Ev\|_2^2 + C \left(1 - \frac{1}{\theta}\right) \|V\|_2^2 + c_Y \gamma_G(\dot{U}) \|\mathbb{V}\|_2^2$$

for some $C > 0$ and $\theta > 0$. Then the coercivity of α in $\mathcal{N}$ is guaranteed independently of $c_Y \gamma_G(\dot{U})$. This suggests that in this case bifurcations cannot occur.

By Proposition 1.17 in a neighborhood of $(\dot{y}, p, y, t) = (0, 0, 0, 0)$ we can solve uniquely

$$\mathcal{F}(\dot{y}, p, y, t) = 0$$

in $\xi = (\dot{y}, p)$ to get

$$\dot{y} = \mathcal{G}(y, t) \quad \text{and} \quad p = \mathcal{H}(y, t)$$

where $\mathcal{G}$ and $\mathcal{H}$ are maps of class C^1 from a neighborhood of $(0, 0)$ in $\mathcal{Z} \times \mathbb{R}$ to $\mathcal{Z}$ and $\mathcal{Q}$ respectively. Then we can apply Theorem B.10 to get locally in time a unique solution y of class C^2 to the Cauchy problem

$$\begin{cases} \dot{y} = \mathcal{G}(y, t) \\ y(0) = 0. \end{cases}$$

Then we set $p = \mathcal{H}(y, t)$ which is of class C^1 . It is clear that $\mathcal{F}(\dot{y}, p, y, t) = 0$. In conclusion we have obtained the following theorem.

Theorem 1.18. There exists $S \in (0, T]$ and unique $y \in C^2([0, S], \mathcal{Z})$ and $p \in C^1([0, S], \mathcal{Q})$ such that (y, p) is a solution to Problem 1.4 with S in place of T .

1.2.6 Bifurcation analysis

I plan to apply the Lyapunov–Schmidt procedure and the Crandall–Rabinowitz theorem to study the bifurcations of the equation

$$\mathcal{F}(\dot{y}, p, y, t) = 0$$

to be solved in $\xi = (\dot{y}, p)$.

1.2.7 Example of constitutive relations

We now give an example of constitutive relations that satisfy the hypotheses. We choose

$$F(\dot{\xi}) = \beta \left[(|\dot{\xi}|^2 + a_F^2)^{1/2} - a_F \right]$$

for some $a_F > 0$ and $\beta \in \mathbb{R}$. This is a regularization of $F(\dot{\xi}) = \beta |\dot{\xi}|$.

Remark 1.19. The parameter β is the dilatancy. The mechanical interpretation is as follows. When $\beta > 0$ then $\dot{\lambda} > 0$ where the material flows plastically. This is the behavior of dilative materials such as dense sand: since the grains are very tightly packed when they slide over each other they space out from each other. On the contrary when $\beta < 0$ then $\dot{\lambda} < 0$ where the material flows plastically. This is the behavior of contractive materials such as loose sand: since the grains are not very tightly packed they tend to compact as they slide over each other.

We choose the usual isotropic elastic and viscoelastic tensors

$$\mathfrak{C} = \lambda_{\mathfrak{C}} I_{d \times d} \otimes I_{d \times d} + 2\mu_{\mathfrak{C}} I_{(d \times d) \times (d \times d)} \quad \text{and} \quad \mathfrak{D} = \lambda_{\mathfrak{D}} I_{d \times d} \otimes I_{d \times d} + 2\mu_{\mathfrak{D}} I_{(d \times d) \times (d \times d)}$$

where

$$d\lambda_{\mathfrak{C}} + 2\mu_{\mathfrak{C}} > 0, \quad \mu_{\mathfrak{C}} > 0 \quad \text{and} \quad d\lambda_{\mathfrak{D}} + 2\mu_{\mathfrak{D}} > 0, \quad \mu_{\mathfrak{D}} > 0.$$

We expect that when the micropressure p grows the friction between the grains of the material increases. When $p \leq 0$ there is a residual cohesion between them. Then we choose

$$Y(p) = Y_0 + H\Phi(p)$$

where

$$\Phi(p) = \begin{cases} 0 & \text{if } p < 0 \\ \frac{p^2}{2p_0^2} & \text{if } p \in [0, p_0] \\ 1 - \frac{p_0^2}{2p^2} & \text{if } p \in (p_0, \infty). \end{cases}$$

with $H > 0$ and $p_0 > 0$. Then we choose

$$g(r) = \frac{1}{m} \left[(r^2 + c^2)^{m/2} - c^m \right], \quad G(\dot{\xi}) = g(|\dot{\xi}|)$$

where $m \in (1, 2)$ and $c > 0$. We have

$$g'(r) = r(r^2 + c^2)^{\frac{m-2}{2}}, \quad G'(\dot{\xi}) = g'(|\dot{\xi}|) \frac{\dot{\xi}}{|\dot{\xi}|} = (|\dot{\xi}|^2 + c^2)^{\frac{m-2}{2}} \dot{\xi}.$$

Remark. When c is close to 0 and m is close to 1 we approximate the classical behavior of a plastic material in which $\dot{\xi} \neq 0$ only when $|\sigma_p|$ reach the yield strength Y .

Lemma. We have

$$G''(\dot{\xi}) = (|\dot{\xi}|^2 + c^2)^{\frac{m-2}{2}} \left[I - (2-m) \frac{1}{|\dot{\xi}|^2 + c^2} \dot{\xi} \otimes \dot{\xi} \right].$$

Moreover the smallest eigenvalue of $G''(\dot{\xi})$ is

$$\gamma_G(\dot{\xi}) = (|\dot{\xi}|^2 + c^2)^{\frac{m-2}{2}} \left(1 - (2-m) \frac{|\dot{\xi}|^2}{|\dot{\xi}|^2 + c^2} \right) > 0.$$

Proof. We have

$$g''(r) = \left(1 - (2 - m) \frac{r^2}{r^2 + c^2}\right) (r^2 + c^2)^{\frac{m-2}{m}}.$$

so that

$$\begin{aligned} G''(\dot{\xi}) &= g''(|\dot{\xi}|) \frac{\dot{\xi}}{|\dot{\xi}|} \otimes \frac{\dot{\xi}}{|\dot{\xi}|} + \frac{g'(|\dot{\xi}|)}{|\dot{\xi}|} \left(I - \frac{\dot{\xi}}{|\dot{\xi}|} \otimes \frac{\dot{\xi}}{|\dot{\xi}|} \right) = \\ &= \frac{g'(|\dot{\xi}|)}{|\dot{\xi}|} I + \left(g''(|\dot{\xi}|) - \frac{g'(|\dot{\xi}|)}{|\dot{\xi}|} \right) \frac{\dot{\xi}}{|\dot{\xi}|} \otimes \frac{\dot{\xi}}{|\dot{\xi}|} = \\ &= (|\dot{\xi}|^2 + c^2)^{\frac{m-2}{2}} \left[I - (2 - m) \frac{|\dot{\xi}|^2}{|\dot{\xi}|^2 + c^2} \frac{\dot{\xi}}{|\dot{\xi}|} \otimes \frac{\dot{\xi}}{|\dot{\xi}|} \right]. \end{aligned}$$

Then notice that $G''(\dot{\xi})$ is of the form $M = \alpha(I - c n \otimes n)$ with $\alpha > 0$, $c \in (0, 1)$ and $n \in \mathbb{R}^m$ with $|n| = 1$. Here m is the dimension of DevSym . We notice that

$$Mn = \alpha(n - cn) = \alpha(1 - c)n$$

and if $v \in \text{span}(n)^\perp$ then

$$Mv = \alpha v.$$

This means that the smallest eigenvalues of M is $\alpha(1 - c)$. The thesis follows. $\square$

1.2.8 Old ideas

Suppose that for $t = 0$ we have

$$\mathcal{F}(0, 0, 0, 0) = 0.$$

Now we have to solve

$$\mathcal{G}_1(a, h) = \mathcal{G}_1(\dot{u}, p, h) = \mathcal{F}(\dot{u}, p, h\dot{u}, h) = 0,$$

where $a = (\dot{u}, p)$. Notice that $\mathcal{G}_1(0, 0) = 0$. If we prove that

$$D_a \mathcal{G}_1(0, 0)$$

is an isomorphism then we find a unique small h_1 and a unique small a_1 such that

$$\mathcal{G}_1(a_1, h_1) = \mathcal{F}(\dot{u}_1, p_1, u_1, t_1) = 0$$

where $u_1 = h_1 \dot{u}_1$ and $t_1 = h_1$. Now we have to solve

$$\mathcal{G}_2(a, h) = \mathcal{G}_2(\dot{u}, p, h) = \mathcal{F}(\dot{u}, p, u_1 + h\dot{u}, t_1 + h) = 0.$$

We notice that

$$\mathcal{G}_2(a_1, 0) = \mathcal{F}(\dot{u}_1, p_1, u_1, t_1) = 0.$$

So if we prove that

$$D_a \mathcal{G}_2(a_1, 0)$$

is a isomorphism we find a unique small h_2 and a unique a_2 near a_1 such that

$$\mathcal{G}_2(a_2, h_2) = \mathcal{F}(\dot{u}_2, p_2, u_2, t_2)$$

where $u_2 = u_1 + h_2 \dot{u}_2$ and $t_2 = t_1 + h_2$. Now suppose that we have obtained $\dot{u}_{n-1}$, p_{n-1} , u_{n-1} and t_{n-1} such that

$$\mathcal{F}(\dot{u}_{n-1}, p_{n-1}, u_{n-1}, t_{n-1}) = 0.$$

We define

$$\mathcal{G}_n(\mathbf{a}, h) = \mathcal{F}(\dot{\mathbf{u}}, \mathbf{p}, \mathbf{u}_{n-1} + h\dot{\mathbf{u}}, t_{n-1} + h).$$

Notice that

$$\mathcal{G}_n(\mathbf{a}_{n-1}, 0) = \mathcal{F}(\dot{\mathbf{u}}_{n-1}, \mathbf{p}_{n-1}, \mathbf{u}_{n-1}, t_{n-1}) = 0.$$

So if we prove that

$$D_{\mathbf{a}}\mathcal{G}_n(\mathbf{a}_{n-1}, 0)$$

is an isomorphism then we find a unique small h_n and a unique $\mathbf{a}_n$ near $\mathbf{a}_{n-1}$ such that

$$\mathcal{G}_n(\mathbf{a}_n, h_n) = \mathcal{F}(\dot{\mathbf{u}}_n, \mathbf{p}_n, \mathbf{u}_n, t_n) = 0$$

where $\mathbf{u}_n = \mathbf{u}_{n-1} + h_n\dot{\mathbf{u}}_n$ and $t_n = t_{n-1} + h_n$.

1.3 Numerical approximation

I plan to study numerically the problem using the FEM. The main branch of the solution will be computed using the Newton method. When the stiffness matrix is approximately singular I will try to generalize the method that I used in nonlinear elasticity to detect the bifurcation points and approximate the various branches of solutions that departs from it.

Chapter 2

Large deformation model

2.1 Formulation

2.1.1 Kinematics

Let $\Omega \subset \mathbb{R}^3$ be the reference configuration of the body. The deformation of the body is a map $\chi: \Omega \rightarrow \mathbb{R}^d$. We set $\mathbf{v} = \dot{\chi}$ and $\mathbf{F} = \text{Grad } \chi$ and $\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1}$. We subdivide $\partial\Omega$ into disjoint parts Γ_D and Γ_N . On Γ_D we impose the kinematical constrain $\chi = \bar{\chi}$. To impose this constrain we add the Lagrange multiplier $\mathbf{r}: \Gamma_D \rightarrow \mathbb{R}^d$. The constrain can be written in weak form as

$$\int_{\Gamma_D} (\chi - \bar{\chi}) \cdot \tilde{\mathbf{r}} dA = 0.$$

The plastic distortion is a map $\mathbf{F}_p: \Omega \rightarrow \text{Lin}^+$. We set $\mathbf{L}_p = \dot{\mathbf{F}}_p \mathbf{F}_p^{-1}$ and we assume that $\mathbf{L}_p = \mathbf{D}_p \in \text{Sym}$. We set $\mathbf{U}_p = \text{dev } \mathbf{D}_p$ and $\mathbf{M}_p = \text{Grad } \mathbf{U}_p \cdot \mathbf{F}_p^{-1}$. The elastic distortion is $\mathbf{F}_e = \mathbf{F}\mathbf{F}_p^{-1}$ and we set $\mathbf{L}_e = \dot{\mathbf{F}}_e \mathbf{F}_e^{-1}$.

Lemma 2.1. It holds $\mathbf{L} = \mathbf{L}_e + \mathbf{F}_e \mathbf{D}_p \mathbf{F}_e^{-1}$.

Proof. We have $\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1} = (\dot{\mathbf{F}}_e \mathbf{F}_p + \mathbf{F}_e \dot{\mathbf{F}}_p) \mathbf{F}_p^{-1} \mathbf{F}_e^{-1} = \dot{\mathbf{F}}_e \mathbf{F}_e^{-1} + \mathbf{F}_e \dot{\mathbf{F}}_p \mathbf{F}_p^{-1} \mathbf{F}_e^{-1} = \mathbf{L}_e + \mathbf{F}_e \mathbf{D}_p \mathbf{F}_e^{-1}$. $\square$

We add the kinematical constrain $\text{tr } \mathbf{D}_p = F(\mathbf{U}_p)$ where $F: \text{DevSym} \rightarrow \mathbb{R}$. To impose this constrain we consider a Lagrange multiplier $p: \Omega \rightarrow \mathbb{R}$. The constrain can be written in weak form as

$$\int_{\Omega} (F(\mathbf{U}_p) - \text{tr } \mathbf{L}_p) \tilde{p} dX = 0.$$

2.1.2 Statics

We choose an internal virtual power of the form

$$\tilde{W} = \int_{\Omega} \mathbf{T} : \tilde{\mathbf{L}}_e dX + \int_{\Omega} \left(\mathbf{T}_p : \tilde{\mathbf{U}}_p - p \text{tr } \tilde{\mathbf{L}}_p + \mathbf{K}_p : \tilde{\mathbf{M}}_p \right) dX.$$

Here $\mathbf{T}$ is the macrostress, $\mathbf{T}_p$ is the microstress which can be assumed deviatoric and $\mathbf{K}_p$ is the microhyperstress. We choose an external virtual power of the form

$$\tilde{P} = \int_{\Omega} \mathbf{f} \cdot \tilde{\mathbf{v}} dX + \int_{\Gamma_N} \mathbf{h} \cdot \tilde{\mathbf{v}} dA + \int_{\Gamma_D} \mathbf{r} \cdot \tilde{\mathbf{v}} dA.$$

Definition 2.2 (Principle of Virtual Powers). Equilibrium holds when $\tilde{W} = \tilde{P}$ for all virtual rates.

To get the local version of the equilibrium condition we apply Lemma 2.1 and we write the internal virtual power as

$$\tilde{W} = \int_{\Omega} \mathbf{T}\mathbf{F}^{-T} : \text{Grad } \tilde{v} dX + \int_{\Omega} \left(-\mathbf{T}_e : \tilde{\mathbf{L}}_p + \mathbf{T}_p : \tilde{\mathbf{U}}_p - p \text{tr } \tilde{\mathbf{L}}_p + \mathbf{K}_p : \tilde{\mathbf{M}}_p \right) dX$$

where $\mathbf{T}_e = \mathbf{F}_e^T \mathbf{T} \mathbf{F}_e^{-T}$ is the Mandel stress. Then we apply the Green theorem

$$\begin{aligned} \tilde{W} = & - \int_{\Omega} \text{Div}(\mathbf{T}\mathbf{F}^{-T}) : \text{Grad } \tilde{v} dX + \int_{\partial\Omega} (\mathbf{T}\mathbf{F}^{-T})\mathbf{N} \cdot \tilde{v} dA + \\ & + \int_{\Omega} \left\{ \left[\mathbf{T}_p - \text{dev } \mathbf{T}_e - \text{Div} \left(\mathbf{K}_p \cdot \mathbf{F}_p^{-T} \right) \right] : \tilde{\mathbf{U}}_p - \left[\frac{1}{d} \text{tr } \mathbf{T}_e + p \right] \text{tr } \tilde{\mathbf{L}}_p \right\} dX + \\ & + \int_{\partial\Omega} \left(\mathbf{K}_p \cdot \mathbf{F}_p^{-T} \mathbf{N} \right) : \tilde{\mathbf{U}}_p dA. \end{aligned}$$

Then the local form of the equilibrium condition is

$$\begin{aligned} \text{Div}(\mathbf{T}\mathbf{F}^{-T}) + \mathbf{f} &= 0 \quad \text{in } \Omega \\ (\mathbf{T}\mathbf{F}^{-T})\mathbf{N} &= \mathbf{r} \quad \text{in } \Gamma_D \\ (\mathbf{T}\mathbf{F}^{-T})\mathbf{N} &= \mathbf{h} \quad \text{in } \Gamma_N \\ \mathbf{T}_p - \text{dev } \mathbf{T}_e - \text{Div} \left(\mathbf{K}_p \cdot \mathbf{F}_p^{-T} \right) &= 0 \quad \text{in } \Omega \\ \frac{1}{d} \text{tr } \mathbf{T}_e + p &= 0 \quad \text{in } \Omega \\ \mathbf{K}_p \cdot \mathbf{F}_p^{-T} \mathbf{N} &= 0 \quad \text{in } \partial\Omega. \end{aligned}$$

2.1.3 Constitutive relations

QUI VA ANCORA COMPLETATO. IN OGNI CASO NON MI ASPETTO DI RIUSCIRE AD ESTENDERE L'ANALISI ANCHE AL MODELLO A GRANDI DEFORMAZIONI. SE AVRÒ TEMPO TRATTERÒ IL PROBLEMA NUMERICAMENTE ESTENDENDO I METODI SVILUPPATI PER IL MODELLO A PICCOLE DEFORMAZIONI IN MODO FORMALE.

Appendix A

Notation

A.1 Tensors

$\text{Lin} = \mathbb{R}^{d \times d}$	$d \times d$ matrices
Lin^+	$d \times d$ matrices with positive determinant
Sym	$d \times d$ symmetric matrices
Dev	$d \times d$ deviatoric matrices
DevSym	$d \times d$ symmetric and deviatoric matrices

Appendix B

Functional analysis

B.1 Linear functional analysis

B.1.1 Other

Proposition B.1. Let $\Omega \subset \mathbb{R}^d$ be open and bounded. Let $p \in [1, \infty]$. Let $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$. Then $u_n \rightarrow u$ in $L^p(\Omega)$.

Proof. Since $u_n \rightharpoonup u$ in $W^{1,p}(\Omega)$ then u_n are bounded in $W^{1,p}(\Omega)$. By Rellich–Kondrakov theorem each subsequence of u_n admits a further subsequence that converges strongly in $L^p(\Omega)$. Since we already know the weak convergence in $W^{1,p}(\Omega)$ then the limit is always u . Then necessarily the whole sequence u_n converges strongly to u in $L^2(\Omega)$. $\square$

B.1.2 Generalized mixed elliptic problems

We start with the following theorem.

Theorem B.2 (Babuška). Let V and Q be reflexive Banach spaces. Let $b: Q \times V \rightarrow \mathbb{R}$ be bilinear form. Let $Z = \{z \in V \mid b(q, z) = 0 \text{ for all } q \in Q\}$ and define $W = V/Z$. Assume that b is continuous so that there exists $M > 0$ such that

$$b(q, v) \leq M \|q\|_Q \|v\|_V \quad \text{for all } (q, v) \in Q \times V,$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, w)}{\|q\|_Q} \geq \beta \|w\|_W$$

for all $w \in W$ and that if $q \in Q$ satisfies

$$b(q, v) = 0 \quad \text{for all } v \in V$$

then $q = 0$. Then for each $f \in Q'$ there exists a unique $w \in W$ such that

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q$$

and

$$\|w\|_W \leq \frac{1}{\beta} \|f\|_{Q'}.$$

Proof. Step 1. Consider the linear operator $B: W \rightarrow Q'$ defined by

$$\langle Bw, q \rangle = b(w, q).$$

We notice that

$$|\langle Bw, q \rangle| = \inf_{z \in Z} |b(w + z, q)| \leq M \|q\|_Q \inf_{z \in Z} \|w + z\|_V = M \|q\|_Q \|w\|_W$$

so that $B \in \mathcal{L}(W, Q')$.

Step 2. We prove that B is injective. Indeed if $w \in W$ satisfies $Bw = 0$ we have

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = 0$$

so that $w = 0$.

Step 3. We prove that $\text{range}(B)$ is closed. Indeed let $f_n = Bw_n$ be such that $f_n \rightarrow f$ in Q . Then f_n is a Cauchy sequence. Now

$$\beta \|w_n - w_m\| \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle B(w_n - w_m), q \rangle}{\|q\|_Q} \leq \|f_n - f_m\|_{Q'}.$$

Then also w_n is a Cauchy sequence in W and $w_n \rightarrow w$ in W for some $w \in W$. Now by continuity of B we have $Bw = f$ so that $f \in \text{range}(B)$.

Step 4. We prove that $\text{range}(B) = Q'$. Indeed let $(B^\top) = \{q \in Q \mid b(q, v) = 0 \text{ for all } v \in V\} = \{0\}$ by hypotheses. On the other hand $(B^\top) = \text{range}(B)^\perp$. Since Q is reflexive $\text{range}(B) = (\text{range}(B)^\perp)^\perp = Q'$.

Step 5. We have proved that B is a bijection from W to Q' . Then for each $f \in Q'$ there exists a unique $w \in W$ such that $Bw = f$. Moreover

$$\beta \|w\|_W \leq \sup_{q \in Q \setminus \{0\}} \frac{\langle Bw, q \rangle}{\|q\|_Q} = \sup_{q \in Q \setminus \{0\}} \frac{\langle f, q \rangle}{\|q\|_Q} \leq \|f\|_{Q'}.$$

This completes the proof. $\square$

Lemma B.3. Assume that the hypotheses of Theorem B.2 holds. Then

$$\sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\|w\|_W} \geq \beta \|q\|_Q$$

for all $q \in Q$.

Proof. For each $f \in Q'$ let $w_f \in W$ the solution to

$$b(q, w) = \langle f, q \rangle \quad \text{for all } q \in Q.$$

Notice that $\|f\|_{Q'} \geq \beta \|w\|_W$. Now

$$\|q\|_Q = \sup_{f \in Q' \setminus \{0\}} \frac{\langle f, q \rangle}{\|f\|_{Q'}} = \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\|f\|_{Q'}} \leq \sup_{f \in Q' \setminus \{0\}} \frac{b(q, w_f)}{\beta \|w_f\|_W} \leq \sup_{w \in W \setminus \{0\}} \frac{b(q, w)}{\beta \|w\|_W}.$$

This completes the proof. $\square$

Then we get immediately the following proposition.

Corollary B.4. Assume the hypotheses of Theorem B.2. Then for each $g \in W'$ there exists a unique $p \in Q$ such that

$$b(p, w) = \langle g, w \rangle \quad \text{for all } w \in W$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{W'}.$$

Proof. Apply first Lemma B.3 and then Theorem B.2 with the role of W and Q swapped. $\square$

Remark B.5. Assume that $g \in V'$ is such that $\langle g, z \rangle = 0$ for all $z \in Z$. Then we can define g as an element of W' . Indeed for each $w \in W$ we take a representative $v \in V$ still denoted with w and we define $\langle g, w \rangle = \langle g, v \rangle$. The definition does not depend on the choice of the representative. Moreover

$$\|g\|_{W'} = \sup_{w \in W \setminus \{0\}} \frac{\langle g, w \rangle}{\|w\|_W} = \sup_{v \in V \setminus \{0\}} \frac{\langle g, v \rangle}{\|v\|_V} = \|g\|_{V'}.$$

where we use that since V is reflexive for each $w \in W$ there exists a representative $v \in V$ such that $\|w\|_W = \|v\|_V$ and that for any other representative $u \in V$ we have $\|w\|_W \leq \|u\|_V$. In this case the Corollary B.4 can be formulated as follows. There exists a unique $p \in Q$ such that

$$b(p, v) = \langle g, v \rangle \quad \text{for all } v \in V$$

and

$$\|p\|_Q \leq \frac{1}{\beta} \|g\|_{V'}.$$

The following theorem is the main result of this Section.

Theorem B.6. Let $V, \tilde{V}, Q$ and $\tilde{Q}$ be reflexive Banach spaces. Let $b: Q \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Let $\tilde{Z} = \{\tilde{z} \in \tilde{V} \mid b(q, \tilde{z}) = 0 \text{ for all } q \in Q\}$ and define $\tilde{W} = \tilde{V}/\tilde{Z}$. Assume that there exists $M > 0$ such that

$$b(q, \tilde{v}) \leq M \|q\|_Q \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } q \in Q \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\beta > 0$ such that

$$\sup_{q \in Q \setminus \{0\}} \frac{b(q, \tilde{w})}{\|q\|_Q} \geq \beta \|\tilde{w}\|_{\tilde{W}} \quad \text{for all } \tilde{w} \in \tilde{W}$$

and that if $q \in Q$ satisfies

$$b(q, \tilde{v}) = 0 \quad \text{for all } \tilde{v} \in \tilde{V}$$

then $q = 0$. Let $c: \tilde{Q} \times V \rightarrow \mathbb{R}$ be a bilinear form. Let $Z = \{z \in V \mid b(\tilde{q}, z) = 0 \text{ for all } \tilde{q} \in \tilde{Q}\}$ and define $W = V/Z$. Assume that there exists $N > 0$ such that

$$c(\tilde{q}, v) \leq N \|\tilde{q}\|_{\tilde{Q}} \|v\|_V \quad \text{for all } \tilde{q} \in \tilde{Q} \text{ and } v \in V.$$

Assume also that there exists $\gamma > 0$ such that

$$\sup_{\tilde{q} \in \tilde{Q} \setminus \{0\}} \frac{c(\tilde{q}, w)}{\|\tilde{q}\|_{\tilde{Q}}} \geq \gamma \|w\|_W \quad \text{for all } w \in W$$

and that if $\tilde{q} \in \tilde{Q}$ satisfies

$$c(\tilde{q}, v) = 0 \quad \text{for all } v \in V$$

then $\tilde{q} = 0$. Let $\alpha: V \times \tilde{V} \rightarrow \mathbb{R}$ be a bilinear form. Assume that there exists a constant L such that

$$\alpha(v, \tilde{v}) \leq L \|v\|_V \|\tilde{v}\|_{\tilde{V}} \quad \text{for all } v \in V \text{ and } \tilde{v} \in \tilde{V}.$$

Assume also that there exists $\alpha > 0$ such that

$$\sup_{\tilde{v} \in \tilde{V} \setminus \{0\}} \frac{\alpha(z, \tilde{v})}{\|\tilde{v}\|_{\tilde{V}}} \geq \alpha \|z\|_V \quad \text{for all } z \in Z$$

and that if $\tilde{z} \in \tilde{Z}$ satisfies

$$\alpha(v, \tilde{z}) = 0 \quad \text{for all } v \in V$$

then $\tilde{z} = 0$. Then for all $f \in \tilde{V}'$ and $g \in \tilde{Q}'$ there exists a unique $(u, p) \in V \times Q$ such that

$$\begin{aligned} \alpha(u, \tilde{v}) + b(p, \tilde{v}) &= \langle f, \tilde{v} \rangle \quad \text{for all } \tilde{v} \in \tilde{V} \\ c(\tilde{q}, u) &= \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}. \end{aligned}$$

Moreover

$$\begin{aligned} \|u\|_V &\leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'}, \\ \|p\|_Q &\leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha}\right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'}. \end{aligned}$$

Proof. By Theorem B.2 there exists a unique $w \in W$ such that

$$c(\tilde{q}, w) = \langle g, \tilde{q} \rangle \quad \text{for all } \tilde{q} \in \tilde{Q}$$

and

$$\|w\|_W \leq \frac{1}{\gamma} \|g\|_{\tilde{Q}'}.$$

We identify w with one of its representative in V and we can choose it such that $\|w\|_V = \|w\|_W$. We notice that

$$\alpha(w, \tilde{v}) \leq L \|w\|_V \|\tilde{v}\|_{\tilde{V}}.$$

Then $\alpha(w, \cdot) \in \tilde{V}' \subset \tilde{Z}'$. Then $\langle f, \cdot \rangle - \alpha(w, \cdot) \in \tilde{Z}'$. Then by Theorem B.2 the equation

$$\alpha(z, \tilde{z}) = \langle f, \tilde{z} \rangle - \alpha(w, \tilde{z}) \quad \text{for all } \tilde{z} \in \tilde{Z}$$

has a unique solution $z \in Z$ that satisfies

$$\|z\|_V \leq \frac{1}{\alpha} \left(\|f\|_{\tilde{V}'} + \frac{L}{\gamma} \|g\|_{\tilde{Q}'} \right).$$

Now define $u = w + z \in V$ and we notice that

$$\alpha(u, \tilde{z}) = \langle f, \tilde{z} \rangle \quad \text{for all } \tilde{z} \in \tilde{Z} \tag{B.1.1}$$

and

$$\|u\|_V \leq \frac{1}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'} + \frac{1}{\alpha} \|f\|_{\tilde{V}'}.$$

Now we notice by (B.1.1) we have

$$\langle f, \tilde{z} \rangle - \alpha(u, \tilde{z}) = \langle f, \tilde{z} \rangle - \alpha(w, \tilde{z}) - \alpha(z, \tilde{z}) = 0 \quad \text{for all } \tilde{z} \in \tilde{Z}.$$

Then by Remark B.5 we have $\langle f, \cdot \rangle - \alpha(u, \cdot) \in \tilde{W}'$ and the equation

$$b(p, \tilde{v}) = \langle f, \tilde{v} \rangle - \alpha(u, \tilde{v}) \quad \text{for all } \tilde{v} \in \tilde{V}$$

has a unique solution $p \in Q$ that satisfies

$$\|p\|_Q \leq \frac{1}{\beta} (\|f\|_{\tilde{V}'} + L \|u\|_V) \leq \frac{1}{\beta} \left(1 + \frac{L}{\alpha}\right) \|f\|_{\tilde{V}'} + \frac{L\beta}{\gamma} \left(1 + \frac{L}{\alpha}\right) \|g\|_{\tilde{Q}'}.$$

This concludes the proof. $\square$

B.2 Nonlinear functional analysis

B.2.1 Other

Lemma B.7. Let X and Y be Banach spaces. Let U an open subset of X and $F: U \rightarrow Y'$. Let $x \in U$. Assume that there exists $A \in \mathcal{L}(X, Y')$ and for all $v \in X$ there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left| \left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle \right| \leq \omega(\epsilon) \|y\|_Y$$

for all $y \in Y$ and $\epsilon \in \mathbb{R}$ sufficiently small. Then F is Gateaux differentiable in x with Gateaux differential A .

Proof. Fix $v \in X$. Then by hypotheses there exists $\omega: \mathbb{R} \rightarrow [0, \infty)$ with $\omega(\epsilon) \rightarrow 0$ as $\epsilon \rightarrow 0$ such that

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} = \sup_{y \in Y \setminus \{0\}} \frac{\left| \left\langle \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av, y \right\rangle \right|}{\|y\|_Y} \leq \omega(\epsilon).$$

Then

$$\left\| \frac{F(x + \epsilon v) - F(x)}{\epsilon} - Av \right\|_{Y'} \rightarrow 0$$

as $\epsilon \rightarrow 0$. This is exactly the thesis. $\square$

Lemma B.8. Let X and Y be two Banach spaces. Let A_n and A in $\mathcal{L}(X, Y')$ such that for all $x \in X$ and $y \in Y$ we have

$$|\langle A_n x - Ax, y \rangle| \leq \omega_n \|x\|_X \|y\|_Y$$

where $\omega_n \rightarrow 0$. Then $A_n \rightarrow A$ in $\mathcal{L}(X, Y')$.

Proof. We have

$$\|A_n - A\|_{\mathcal{L}(X, Y')} = \sup_{x \in X \setminus \{0\}} \frac{\|A_n x - Ax\|_{Y'}}{\|x\|_X} = \sup_{x \in X \setminus \{0\}} \sup_{y \in Y \setminus \{0\}} \frac{|\langle A_n x - Ax, y \rangle|}{\|x\|_X \|y\|_Y} \leq \omega_n \rightarrow 0.$$

This concludes the proof. $\square$

B.2.2 Implicit function theorem

Theorem B.9. Let X, Y and E be Banach spaces. Let U and open set of $X \times Y$ and $(x_0, y_0) \in U$. Let $F \in C^1(U, E)$ such that $F(x_0, y_0) = 0$ and $D_x F(x_0, y_0) \in \text{Iso}(X, E)$. Then we can find $r > 0$ and $s > 0$ with $B(x_0, r) \times B(y_0, s) \subset U$ such that there exist a unique $\chi \in C^1(B(y_0, s), B(x_0, r))$ satisfying

$$F(\chi(y), y) = 0 \quad \text{for all } y \in B(y_0, s).$$

Moreover if F is of class C^r with $r \geq 2$ also χ is of class C^r .

B.2.3 Local existence theorem for Cauchy problems in Banach spaces

Theorem B.10. Let X be a Banach space and let U be an open subset of $X \times \mathbb{R}$ with $(x_0, 0) \in U$. Let $G \in C^1(U, X)$. Then there exists $T > 0$ such that the Cauchy problem

$$\begin{cases} \dot{x}(t) = G(x(t), t) & \text{for } t \in [0, T] \\ x(0) = x_0 \end{cases}$$

admits a unique solution $x \in C^2([0, T], X)$.

Proof. Since G is of class C^1 there exists a neighborhood $V \subset U$ of $(x_0, 0)$ such that G and $D_x G$ are bounded in V . Let $M = \sup_V \|G\|_X < \infty$ and $L = \sup_V \|D_x G\|_{\mathcal{L}(X, X)}$. We notice that G is Lipschitz continuous with respect to x and uniformly in t on V with Lipschitz constant L . Let $B = B(x_0, r)$ for some $r > 0$ and $T > 0$ such that $B(x_0, r) \times (-T, T) \subset V$. We introduce the normed space $\mathcal{X} = C^0([0, T], B)$ endowed with the norm $\|x\|_{\mathcal{X}} = \sup_{t \in [0, T]} \|x(t)\|_X$. Since $\mathcal{X}$ is a closed subset of $C^0([0, T], B)$ it is a complete metric space. Now introduce the map

$$T: B \rightarrow C^0([0, T], B)$$

defined by

$$T(x)(t) = x_0 + \int_0^t G(x(s), s) ds.$$

We notice that $\|T(x) - x_0\|_{\mathcal{X}} \leq TM$. Up to choose a smaller T we can assume that $TM < r$ so that actually $T: B \rightarrow B$. Moreover for any x_1 and x_2 in $\mathcal{X}$ we have

$$\|T(x_2) - T(x_1)\|_{\mathcal{X}} \leq \int_0^T \|G(x_2(s), s) - G(x_1(s), s)\|_X ds \leq TL \|x_2 - x_1\|_{\mathcal{X}}.$$

Up to choose a smaller T we can assume that $TL < 1$. In this case T is a strict contraction. By the Banach–Caccioppoli fixed point theorem there exists a unique $x \in B$ such that $Tx = x$. Since Tx is of class C^2 then also x is of class C^2 . $\square$

B.2.4 Fixed point theorems

Theorem B.11 (Schauder fixed point theorem). Let X be a Banach space and $C \subset X$ closed and convex. Let $F: C \rightarrow C$ continuous and with compact range. Then F admits a fixed point.

B.2.5 Browder–Minty theorem

Let X be a Banach space and $T: X \rightarrow X'$ be a nonlinear operator and $f \in X'$. The Browder–Minty theorem allows to study the existence of solutions to the equation $T(x) = f$ when there is not any variational formulation. The key assumption is that T is coercive and monotone, a property that makes T similar in some sense to an elliptic operator.

Definition B.12. Let X be a Banach space and let $T: X \rightarrow X'$. We say that

- (i) T is bounded if maps bounded sets of X into bounded sets of X' ,
- (ii) T is coercive if

$$\frac{\langle T(x), x \rangle}{\|x\|} \rightarrow \infty \quad \text{as } \|x\| \rightarrow \infty$$

and that

(iii) T is monotone if

$$\langle T(x) - T(y), x - y \rangle \geq 0 \quad \text{for all } x \text{ and } y \text{ in } X.$$

We now prove a lemma that establish the equivalence between the equation $T(x) = f$ and a variational inequality under the assumption that T is monotone.

Lemma B.13. Let $T: X \rightarrow X'$ be continuous and $f \in X'$. Then if $x \in X$ is a solution to the variational inequality

$$\langle T(y) - f, y - x \rangle \geq 0 \quad \text{for all } y \in X$$

it is also a solution to the equation $T(x) = f$. Moreover if T is monotone then any solution to the equation $T(x) = f$ is also a solution to the variational inequality.

Proof. Let $y = x + tv$ with $t > 0$ and $v \in X$. Then if $x \in X$ is a solution to the variational inequality we have

$$\langle T(x + tv) - f, v \rangle \geq 0.$$

Letting $t \rightarrow 0$ we have $\langle T(x) - f, v \rangle \geq 0$. By arbitrariness of v we have $T(x) = f$. Conversely let $x \in X$ be such that $T(x) = f$. By monotonicity

$$\langle T(y) - f, y - x \rangle = \langle T(y) - T(x), y - x \rangle \geq 0.$$

Then x is a solution to the variational inequality. $\square$

Now we study the equation $T(x) = f$ when the Banach space is finite dimensional. This results will be used to get the general result using the Galerkin method.

Proposition B.14. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous and coercive. Then for every $f \in \mathbb{R}^n$ the equation $T(x) = f$ has at least one solution.

Proof. Without loss of generality we can assume that $f = 0$. Indeed otherwise define $S(x) = T(x) - f$ and notice that S is still continuous and coercive. Indeed

$$\frac{S(x) \cdot x}{|x|} = \frac{T(x) \cdot x}{|x|} - \frac{f \cdot x}{|x|} \geq \frac{T(x) \cdot x}{|x|} - |f|.$$

Now define $G: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $G(x) = x - T(x)$. We notice that

$$G(x) \cdot x = |x|^2 - T(x) \cdot x.$$

Since T is coercive there exists an $r > 0$ such that $|x|^2 - G(x) \cdot x = T(x) \cdot x > 0$ for all $x \in \mathbb{R} \setminus B(0, r)$. We define $\Pi: \mathbb{R}^n \rightarrow C$ with $C = \overline{B}(0, r)$ by

$$\Pi(x) = \begin{cases} x & \text{if } |x| \leq r \\ r \frac{x}{|x|} & \text{if } |x| > r. \end{cases}$$

We set $H(x) = \Pi(G(x))$ and we notice that $H: C \rightarrow C$ and it is continuous. By Brouwer fixed point theorem there exists $x \in C$ such that $H(x) = x$. If $x \in \overset{\circ}{C}$ then $x = H(x) = \Pi(G(x)) = G(x)$ so that $T(x) = 0$. If we prove that $x \notin \partial C$ we have concluded. By contradiction assume that $|x| = r$. Then $|G(x)| \geq r$ and

$$|x|^2 = x \cdot x = H(x) \cdot x = \frac{r}{|G(x)|} G(x) \cdot x < \frac{r}{|G(x)|} |x|^2 \leq |x|^2$$

where we used that $|x|^2 > G(x) \cdot x$ for $|x| \geq r$. This is a contradiction. $\square$

We are now in position to prove the Browder–Minty theorem.

Theorem B.15. Let X be a reflexive and separable Banach space. Let $T: X \rightarrow X'$ be bounded, continuous, coercive and monotone. Then for every $f \in X'$ there exists a solution $x \in X$ to the equation $T(x) = f$.

Proof. As anticipated the proof relies on the Galerkin method.

Step 1. Let $\{y_n\} \subset X$ such that the linear combinations of y_n are dense in X . Let

$$X_n = \text{span} \{y_m\}_{m=1}^n.$$

We define $f_n \in X'_n$ by $\langle f_n, x \rangle = \langle f, x \rangle$ for all $x \in X_n$. We also define $T_n: X_n \rightarrow X'_n$ by $\langle T_n(x), y \rangle = \langle T(x), y \rangle$ for all x and y in X_n .

Step 2. We prove that T_n are still bounded, continuous and coercive. For all $x \in X_n$ we have

$$\|T_n(x)\|_{X'_n} = \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x), y \rangle}{\|y\|_X} = \|T(x)\|_X$$

so the boundedness of T_n follows from the boundedness of T . Moreover let $x_k \rightarrow x$ in X_n . Then

$$\begin{aligned} \|T_n(x_k) - T_n(x)\|_{X'_n} &= \sup_{y \in X_n \setminus \{0\}} \frac{\langle T_n(x_k) - T_n(x), y \rangle}{\|y\|_{X_n}} \leq \sup_{y \in X \setminus \{0\}} \frac{\langle T(x_k) - T(x), y \rangle}{\|y\|_X} = \\ &= \|T(x_k) - T(x)\|_{X'} \rightarrow 0 \end{aligned}$$

and this proves that T_n is continuous. Moreover for $x \in X_n$

$$\frac{\langle T_n(x), x \rangle}{\|x\|_{X_n}} = \frac{\langle T(x), x \rangle}{\|x\|_X}$$

so the coercivity of T_n follows immediately from the coercivity of T . By B.14 we have that for each n there exists $x_n \in X_n$ that solves $T_n(x_n) = f_n$.

Step 3. We have the estimate

$$\frac{\langle T(x_n), x_n \rangle}{\|x_n\|_X} \leq \frac{\langle T_n(x_n), x_n \rangle}{\|x_n\|_X} = \frac{\langle f_n, x_n \rangle}{\|x_n\|_X} = \frac{\langle f, x_n \rangle}{\|x_n\|_X} \leq \|f\|.$$

Since T is coercive then x_n are bounded in X and since T is bounded then $T(x_n)$ are bounded in X' . Then up to a subsequence

$$\begin{aligned} x_n &\rightharpoonup x \quad \text{in } X \\ T(x_n) &\overset{*}{\rightharpoonup} g \quad \text{in } X' \end{aligned}$$

for some $x \in X$ and $g \in X'$.

Step 4. We first prove that $g = f$. To this aim we fix any y_m and we notice that

$$\langle g - f, y_m \rangle = \lim_{n \rightarrow \infty} \langle T(x_n) - f, y_m \rangle = 0.$$

Then by density we have indeed $g = f$. Next we have

$$\langle T(x_n), x_n \rangle = \langle T_n(x_n), x_n \rangle = \langle g_n, x_n \rangle = \langle g, x_n \rangle \rightarrow \langle g, x \rangle.$$

Step 5. Eventually using the monotonicity of T we have

$$0 \geq \langle T(y) - T(x_n), y - x_n \rangle \rightarrow \langle T(y) - g, y - x \rangle = \langle T(y) - f, y - x \rangle.$$

Applying Lemma B.13 we get the thesis. $\square$

Actually to obtain the same existence result the monotonicity assumption can be weakened.

Definition B.16. Let X be a Banach space and let $T: X \rightarrow X'$. We say that T is pseudomonotone if it is bounded and every time

$$x_k \rightharpoonup x \text{ in } X \quad \text{and} \quad \limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0$$

then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$.

Theorem B.17. Let X be a separable and reflexive Banach space. Let $T: X \rightarrow X'$ be continuous, coercive and pseudomonotone. Then for every $f \in X'$ there exists $x \in X$ such that $T(x) = f$.

Proof. The proof is exactly the same as the one of Theorem B.15 except to Step 5. We know that $x_k \rightharpoonup x$ in X , that $T(x_k) \xrightarrow{*} f$ in X' and $\langle T(x_k), x_k \rangle \rightarrow \langle f, x \rangle$. Then in particular

$$\langle T(x_k), x_k - x \rangle = \langle T(x_k), x_k \rangle - \langle T(x_k), x \rangle \rightarrow \langle f, x \rangle - \langle f, x \rangle = 0.$$

Since T is pseudomonotone then

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. But

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \langle f, x - y \rangle.$$

Then

$$\langle f, x - y \rangle \geq \langle T(x), x - y \rangle$$

for all $y \in X$. By arbitrariness of y we have $T(x) = f$. □

The following proposition establish the relationship between monotone and pseudomonotone operators.

Proposition B.18. Let X be a Banach space. Let $T: X \rightarrow X'$ be continuous, bounded and monotone. Then T is pseudomonotone.

Proof. Let $x_k \rightharpoonup x$ with

$$\limsup_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \leq 0.$$

Since T is monotone then

$$\langle T(x_k), x_k - x \rangle \geq \langle T(x), x_k - x \rangle \rightarrow 0.$$

This implies also

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle \geq 0.$$

Then

$$\lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle = 0.$$

Now let $y \in X$ and set $y_\epsilon = (1 - \epsilon)x + \epsilon y$ for $\epsilon > 0$. By monotonicity

$$0 \leq \langle T(x_k) - T(y_\epsilon), x_k - y_\epsilon \rangle = \langle T(x_k) - T(y_\epsilon), \epsilon(x - y) + x_k - x \rangle.$$

Rearranging the terms we have

$$\epsilon \langle T(x_k), x - y \rangle \geq \langle T(y_\epsilon), x_k - x \rangle - \langle T(x_k), x_k - x \rangle + \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Passing to the limit $k \rightarrow \infty$ with $\epsilon > 0$ fixed

$$\epsilon \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \epsilon \langle T(y_\epsilon), x - y \rangle.$$

Dividing by ϵ and passing to the limit $\epsilon \rightarrow 0^+$ we obtain using the continuity of T that

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq \langle T(x), x - y \rangle.$$

Eventually

$$\liminf_{k \rightarrow \infty} \langle T(x_k), x_k - y \rangle = \lim_{k \rightarrow \infty} \langle T(x_k), x_k - x \rangle + \liminf_{k \rightarrow \infty} \langle T(x_k), x - y \rangle \geq 0.$$

This proves that T is pseudomonotone. $\square$

The following result is useful.

Proposition B.19. Let X be a Banach space. Let T_1 and T_2 from X into X' be pseudomonotone operators. Then also $T_1 + T_2$ is pseudomonotone.

Proof. Let $x_k \rightarrow x$ with

$$\limsup_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - x \rangle \leq 0.$$

We claim that

$$\limsup_{k \rightarrow \infty} \langle T_i(x_k), x_k - x \rangle \leq 0 \tag{B.2.1}$$

for both $i = 1$ and $i = 2$. By contradiction

$$\limsup_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Up to taking a subsequence we can assume that

$$\lim_{k \rightarrow \infty} \langle T_2(x_k), x_k - x \rangle = \epsilon > 0.$$

Then

$$\limsup_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \leq -\epsilon < 0. \tag{B.2.2}$$

Then since T_1 is pseudomonotone

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k), x_k - x \rangle \geq 0$$

but this contradicts (B.2.2). Then (B.2.1) holds. Since T_i is pseudomonotone then for all $y \in X$ we have

$$\liminf_{k \rightarrow \infty} \langle T_i(x_k), x_k - y \rangle \geq \langle T_i(x), x - y \rangle.$$

It follows that

$$\liminf_{k \rightarrow \infty} \langle T_1(x_k) + T_2(x_k), x_k - y \rangle \geq \langle T_1(x) + T_2(x), x - y \rangle.$$

This means that $T_1 + T_2$ is pseudomonotone. $\square$

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